

Infinitely many closed geodesic images on every Riemannian sphere

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Abstract

We resolve the sphere case of the closed-geodesic infinitude problem: every smooth Riemannian metric on the standard sphere S^n , $n \geq 2$, has infinitely many prime closed geodesics with pairwise distinct images. The two-sphere case is classical; the result in higher dimensions includes degenerate metrics. Using finite covers and theorems of Perelman and Rademacher–Taimanov, we also obtain the same conclusion for every smooth Riemannian metric on a nonempty closed smooth three-manifold.

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1 Introduction

The closed-geodesic infinitude problem asks whether every closed Riemannian manifold of dimension at least two has infinitely many geometrically distinct closed geodesics. A closed geodesic is *prime* if it traverses its periodic trajectory once. Geometric distinction means distinction of images: changing the initial point, reversing orientation, or taking an iterate produces no new image. A prime geodesic may have self-intersections. We resolve the sphere case of the infinitude problem.

Theorem 1.1. *For every integer $n \geq 2$ and every smooth Riemannian metric on the standard smooth sphere S^n , there are infinitely many prime closed geodesics with pairwise different images.*

Birkhoff developed a minimax construction for closed geodesics on surfaces of genus zero [4, Section 17]. The existence of a nonconstant closed geodesic on every positive-dimensional closed Riemannian manifold goes back to Lyusternik and Fet [23]. Infinitude requires separating new geodesics from repeated traversals of old ones. Bott’s iteration theory [6] and the local Morse theory of Gromoll and Meyer [16] make that distinction quantitative. In particular, for a simply connected closed manifold, unbounded Betti numbers of the free loop space over a field force infinitely many geometrically distinct closed geodesics. Vigué-Poirrier and Sullivan [28] showed that rational cohomology requiring at least two algebra generators supplies this topological growth. Spheres have one-generator cohomology and bounded degreewise loop homology, so this criterion does not settle their infinitude problem.

The two-sphere case is classical. It follows from Bangert’s work together with that of Franks or Hingston [3, 14, 18]. De Philippis, Marini, Mazzucchelli and Suhr extended this result to reversible Finsler two-spheres [13, Theorem 1.5]. Albach’s more recent work gives a growth exponent of at least two for the number of geodesic images of bounded length in that reversible Finsler setting [1, Theorem 1.1].

In higher dimensions, Rademacher proved infinitude for strongly bumpy metrics on compact simply connected manifolds and proved that such metrics are generic [26, Theorems 1.1–1.3]. Finite multiplicity results also hold without genericity: Long and Duan proved

the existence of at least two geometrically distinct closed geodesics on every Riemannian three-sphere [22]. These results do not establish infinitude for arbitrary degenerate metrics; a limiting argument from generic metrics need not preserve distinct prime images. For manifolds with positive first Betti number, Contreras and Mazzucchelli recently proved infinitude for generic metrics [12, Corollary B]. Our higher-dimensional argument addresses the unrestricted metric problem.

Earlier infinitude claims also require care. Asselle and Mazzucchelli [2, Introduction] identify the proof in Klingenberg’s monograph [21] as erroneous. Charles [7, Theorem 6.14] later claimed infinitude on every closed Riemannian manifold. We identify two unsupported steps in that argument: the boundary-rank equality in Equation (53) requires maximal ranks not established there, and the constant-ball modification in Equation (54) does not provide the nondegenerate Morse critical points used in the conclusion. These are objections to the proofs, not counterexamples to their conclusions.

Corollary 1.2. *Let N be a closed manifold admitting a finite smooth cover diffeomorphic to a standard sphere S^n , where $n \geq 2$. Every smooth Riemannian metric on N has infinitely many geometrically distinct prime closed geodesics.*

This includes real projective spaces and smooth spherical space forms. The passage between a manifold and its finite covers is classical [28, pp. 633–634]. Section 11 proves the stated form by showing that projection has finite fibers on prime geodesic images, including self-intersections and nonregular covers. The conclusion concerns the number of images. Even on a round sphere, all prime geodesics have the same length.

The finite-cover corollary also combines with a theorem of Rademacher and Taimanov to settle the infinitude problem in dimension three.

Corollary 1.3 (Closed three-manifolds). *Let M be a nonempty closed smooth three-manifold, not necessarily orientable. Every smooth Riemannian metric on M has infinitely many prime closed geodesics with pairwise distinct images.*

Proof. It suffices to work on one connected component, so assume that M is connected. If $\pi_1(M)$ is infinite, Rademacher and Taimanov [27, Theorem 2] give infinitely many geometrically distinct closed geodesics for the given metric. Their Riemannian convention identifies geodesics exactly when their images coincide [27, Section 3]. Taking the least positive period gives a prime representative with the same image. Their theorem includes the nonorientable case.

If $\pi_1(M)$ is finite, the elliptization case of Perelman’s geometrization theorem says that M is diffeomorphic to a spherical space form S^3/Γ , where Γ is finite and acts freely and smoothly on S^3 [24, Corollary 0.2(b)]. Composing the quotient map with a diffeomorphism $S^3/\Gamma \rightarrow M$ gives a finite smooth cover $S^3 \rightarrow M$. Corollary 1.2 therefore applies. Only the smooth covering is used, so the given Riemannian metric remains arbitrary. $\square$

This corollary uses only the infinitude conclusion from Rademacher–Taimanov. It carries no quantitative length-counting assertion for the finite-fundamental-group case.

1.1 Idea of the proof

Assume that a metric on S^n , $n \geq 3$, has only finitely many prime geodesic images. Every nonconstant critical loop is then an iterate of one of finitely many oriented trajectories. Their local homology has uniformly bounded rank and lies near finitely many lines relating index to length. The difficulty is that sublevel homology can also contain classes that disappear at a later level. We first control these temporary classes and then use the symmetry of double loops to obtain a contradiction.

The ordinary homological input is a uniform filling statement for every metric, independent of the finite-prime assumption. Square-root energy is the L^2 -norm of the velocity of a period-one loop. There is a constant D such that a cycle below square-root energy a which bounds in the full loop space already bounds below $a + D$, in every degree (Theorem 6.1). Fixed-degree shortening first gives a bound that may depend on degree. A cut operation paired with a cohomology power lowers a high degree into a fixed finite range, and a loop product restores it. The resonance theorem of Hingston and Rademacher [19] makes the lengths removed and added differ by a bounded amount.

The operations underlying this step come from Chas and Sullivan’s loop product [8] and Goresky and Hingston’s relative cohomology product and its length filtration [15]. Cohen and Jones gave a Thom-spectrum realization of the loop product and its unit [11]. Quantitative shortening of loop families has a related history in the work of Nabutovsky and Rotman [25]; the fixed-degree additive filling estimate of Block, Manin and Weinberger [5, Theorem 5.1] is a direct predecessor of the first step here. We prove that step in the strong H^1 topology before removing its degree dependence. Recent work of Cieliebak, Oancea and Shelukhin extends resonance to further compact rank-one symmetric spaces under dimension and coefficient restrictions, and to finite sphere quotients with local coefficients [10, Theorem 6.1 and Corollary 8.4].

The essential comparison is the filtered cut-join identity of Theorem 5.4. Its chain constructions build on the string-topology models of Hingston and Wahl [20]. The comparison uses an absolute cut with a diagonal-relative Thom cocycle; its value on a suitable power is the constant-loop fundamental class, the unit for the loop product (Proposition 5.6). We prove both the identity and this unit statement, including their filtration bounds.

The remaining argument uses reflection of loops and translation by half a period. With coefficients in $\mathbb{F}_2$, let B_a denote reflection-equivariant cohomology below square-root energy a , relative to constant loops, and let B_∞ denote the corresponding global group. Define

$$T(a) = \dim \operatorname{coker}(B_\infty \longrightarrow B_a),$$

where the dimension is summed over all degrees. Thus $T(a)$ counts the extra reflection classes at that sublevel. Theorem 9.1 shows that $T(a)$ is uniformly bounded under the finite-prime assumption. The proof compares two filtrations on the half-translation fixed locus: the actual equivariant degree and the degree of its half-length root. An exact-sequence calculation bounds their total discrepancy by $O(a)$ (Theorem 8.2). A long string of extra classes would give $\Omega(a^2)$ discrepancy. The ordinary filling estimate excludes the remaining possibilities.

Finally, the full orthogonal-group action gives

$$T(2a) \geq T(a).$$

At each stage of the round-metric filtration, the global basis has a lower-degree family, called the bottoms, and a second family obtained by cup product with the sphere class pulled back by evaluation at time zero. The equality case requires every nonzero bottom restriction from an odd round-filtration stage to have a partner whose restriction is independent modulo all bottom restrictions in its degree (Theorem 10.2). Because T is a bounded integer, an interval where it attains its maximum propagates to arbitrarily large intervals under doubling. A bottom generator must first appear at some crossing in such an interval. Evaluation cup vanishes on that local crossing, so its partner is still zero. This contradicts the equality case.

1.2 Organization and reusable ingredients

Section 2 fixes the loop-space, coefficient and symmetry conventions. Section 3 establishes the local crossing theorem with one common flow defining all restriction maps. Its rank proof

reduces arbitrary degeneracy to finitely many root types; it also includes primes of zero mean index. Sections 4–6 prove uniform ordinary filling. Section 7 computes the global generators and their lifts by natural Gysin sequences and the round filtration. Sections 8–10 give the finite-cut estimate, the transient bound and the contradiction.

Two features are useful beyond the final rank argument. The filtered cut–join comparison turns fixed-degree filling into a bound independent of degree. The finite-cut comparison controls actual restriction images before localization has become an isomorphism. These statements are isolated with their assumptions and outputs so that their uses do not depend on the surrounding contradiction argument.

2 Loop spaces, actions and sublevels

We fix notation for the geometric filtration and the three symmetry groups. Except for the use of the classical two-sphere theorem, assume $n \geq 3$ and fix a smooth Riemannian metric g on S^n . Put $d = n - 1$ and

$$\Lambda = H^1(\mathbb{R}/\mathbb{Z}, S^n), \quad M = \{\text{constant loops}\} \cong S^n.$$

We use the strong H^1 topology. The energy, its square root, and length are

$$E(\gamma) = \int_0^1 |\dot{\gamma}(t)|_g^2 dt, \quad F(\gamma) = \sqrt{E(\gamma)}, \quad \ell(\gamma) = \int_0^1 |\dot{\gamma}(t)|_g dt.$$

Then $\ell \leq F$, with equality for constant-speed loops. Write

$$X_a = \{F < a\}, \quad \mathcal{A}_a = \{\ell < a\} \quad (a \in \mathbb{R}).$$

Both spaces are empty for $a \leq 0$; pairs relative to M are used only for $a > 0$. A positive cut a is *regular* if a^2 is a regular value of E . At regular cuts we freely use the corresponding closed finite-dimensional sublevel models, through the homotopy equivalences constructed in Section 3. All maps between sublevels are the maps induced by inclusion.

The orthogonal group $G = O(2)$ acts by changing the loop parameter. We single out reflection r and half-translation τ :

$$(r\gamma)(t) = \gamma(-t), \quad (\tau\gamma)(t) = \gamma(t + \tfrac{1}{2}), \quad R = \langle r \rangle, \quad K = \langle r, \tau \rangle.$$

Thus $K \cong C_2 \times C_2$. The fixed locus of τ consists of double loops. The map

$$\delta : \Lambda \longrightarrow \Lambda^\tau, \quad (\delta\gamma)(t) = \gamma(2t),$$

is an R -equivariant homeomorphism, and $F(\delta\gamma) = 2F(\gamma)$. In particular, $(X_{2a})^\tau$ is the double of X_a .

All homology and cohomology have coefficients $\mathbb{F}_2$. For an H -invariant pair (Y, Z) , let

$$H_H^j(Y, Z) = H^j(EH \times_H Y, EH \times_H Z; \mathbb{F}_2).$$

Ordinary groups have no group subscript. Negative-degree groups are zero, and a total graded group always means the direct sum of its homogeneous components. We use

$$B_a^j = H_R^j(X_a, M), \quad B_\infty^j = H_R^j(\Lambda, M).$$

The determinant character of G defines $w \in H^1(BG; \mathbb{F}_2)$, with the same notation for its restrictions. Let $U \in H^2(BG; \mathbb{F}_2)$ be the mod-two Euler class of the standard plane. On

K , let v be the degree-one character which is nonzero on τ and zero on r . Proposition 7.1 establishes the ring formulas and the natural sequences used below.

Evaluation at zero is R -equivariant for the trivial action on S^n . If $\sigma \in H^n(S^n; \mathbb{F}_2)$ is its nonzero class, write

$$e = \text{ev}_0^* \sigma \in H_R^n(\Lambda; \mathbb{F}_2).$$

Cup product by e acts on every relative reflection group. This operation will distinguish a global pair of classes whose two members cannot first appear at the same isolated critical crossing.

3 Local control without nondegeneracy

We establish uniform bounds for the local groups of all iterates of a finite collection of prime geodesics. The bounds include geodesics of zero mean index. We also retain the actual restriction maps to the half-translation fixed locus and show that evaluation cup product vanishes at each reflection crossing. The analytic preliminaries below apply to every smooth Riemannian metric, independently of the finite-geodesic hypothesis.

3.1 Energy sublevels and finite-dimensional models

Lemma 3.1 (Analytic foundations). *The energy E is smooth on $\Lambda = H^1(S^1, S^n)$, satisfies the Palais–Smale condition, and has precisely the smooth closed geodesics and the constants as critical points. There is a complete strong Riemannian metric on Λ , invariant under $G = O(2)$, for which a bounded normalized negative gradient of E has a complete equivariant flow. Sufficiently small energy sublevels retract equivariantly to M .*

Proof. Embed the sphere in $\mathbb{R}^{n+1}$. Pointwise exponential charts, the embedding $H^1(S^1) \subset C^0(S^1)$, and the one-dimensional H^1 product rule make Λ a smooth Hilbert submanifold of $H^1(S^1, \mathbb{R}^{n+1})$. Its induced strong metric is complete. Indeed, an intrinsic Cauchy sequence is ambient Cauchy; its limit still takes values in the closed sphere, and a local Hilbert chart near that limit bounds intrinsic path distances by ambient distances. Time translations and reflection are isometries. Each acts smoothly, and their action is jointly continuous in the strong H^1 topology.

Extend g to a smooth positive symmetric matrix $A(x)$ along the ambient tangent bundle, preserving the tangent–normal splitting. Then

$$E(\gamma) = \int_0^1 \langle A(\gamma) \dot{\gamma}, \dot{\gamma} \rangle \, dt$$

is smooth. To check compactness, suppose that $E(\gamma_r)$ is bounded and $\|dE(\gamma_r)\| \rightarrow 0$. Pass to a subsequence converging weakly in H^1 and uniformly to γ . Let P_x denote Euclidean tangent projection. The tangent fields

$$\eta_r = P_{\gamma_r}(\gamma_r - \gamma)$$

are bounded in H^1 . Testing $dE(\gamma_r)$ on η_r , every term containing $\gamma_r - \gamma$ without a derivative tends to zero: its uniform norm tends to zero, while the squared velocities have bounded integrals. This includes the terms from differentiating $A(\gamma_r)$ and P_{γ_r} . Consequently

$$\int_0^1 \langle A(\gamma_r) \dot{\gamma}_r, \dot{\gamma}_r - \dot{\gamma} \rangle \, dt \longrightarrow 0.$$

Uniform convergence of the coefficients and weak convergence of the velocities now give convergence of their energy norms. Uniform positivity of A proves strong H^1 convergence.

At a critical point the weak Euler–Lagrange equation says, in a coordinate interval, that the distributional derivative of the metric momentum is an L^1 quadratic expression in the velocity. The momentum, hence the velocity, is continuous. The geodesic equation then gives smoothness. Its periodic boundary condition follows from the first variation. Conversely, geodesics and constants are critical.

Normalize the negative gradient by

$$Z = -\frac{\nabla E}{\sqrt{1 + \|\nabla E\|^2}}.$$

This is a smooth invariant vector field of norm at most one, strict away from the critical set. A bounded vector field on a complete strong Riemannian manifold has a complete flow: a trajectory of finite time has finite length and a limit, through which the local flow extends it.

Finally, near the constants use coordinates $\gamma(t) = \exp_x \xi(t)$, where $\xi \in H^1(S^1, T_x M)$ has mean zero. The inverse function theorem gives a uniform tube over the compact zero section. The normal Hessian of energy is positive coercive by the mean-zero Poincaré inequality. Taylor integration and the positive square-root change of normal coordinates make the energy an exact positive quadratic form there. These constructions are equivariant. Small energy forces both the diameter and the ambient L^2 derivative norm of a loop to be small, so all sufficiently small sublevels lie in this tube. Radial contraction gives the claim. $\square$

Lemma 3.2 (Regular strips). *If a closed bounded energy interval contains no critical value, the gradient flow gives equivariant sublevel equivalences across it. At regular cuts, open and closed sublevel pairs compute the same equivariant cohomology, naturally for inclusions and fixed loci.*

Proof. Palais–Smale implies a positive lower bound on $\|\nabla E\|$ in such an interval. The normalized flow therefore decreases energy at a uniformly positive rate there. A fixed sufficiently long time moves its upper sublevel into its lower sublevel, and the intervening flow gives the required homotopies. A small regular collar at an endpoint similarly compares the open and closed sublevels. Apply these constructions simultaneously at the endpoints of a pair. All maps are equivariant because they use the same invariant flow. $\square$

We shall use closed regular polygon sublevels in the local constructions. Lemma 3.2 identifies their crossing groups with those of the open sublevels $X_a = \{F < a\}$ used elsewhere. Here and below, a cut for polygon energy is the square of the corresponding cut for $F = \sqrt{E}$.

Lemma 3.3 (Polygon models). *Fix $B < \infty$. For every sufficiently large integer N , the energy sublevel $\{E \leq B^2\}$ retracts onto its broken-geodesic polygons with N equal time cells. The retraction preserves every smaller energy sublevel. It commutes with reflection and all time shifts preserving the vertex grid, hence with K when N is even. Regular polygon sublevels through this bound are compact smooth manifolds with boundary. At every critical point in the band, polygon and Hilbert indices and nullities agree. Regular energy cuts are dense in the band, and their sublevels have finite CW models.*

Proof. Choose $\rho > 0$ below the injectivity radius. The open manifold

$$P_N = \{(x_0, \dots, x_{N-1}) \in M^N : d_g(x_j, x_{j+1}) < \rho\}$$

is identified with the polygons formed by minimizing constant-speed segments; subscripts are cyclic. Its energy is

$$f_N = N \sum_{j=0}^{N-1} d_g(x_j, x_{j+1})^2.$$

Choose N so large that $B/\sqrt{N} < \rho$. On each time cell, replace the central subinterval of proportion s , $0 \leq s \leq 1$, by its minimizing segment. Cauchy–Schwarz makes all endpoint distances shorter than ρ , and shows that this replacement does not increase energy. At $s = 1$ the loop is a polygon, and polygons are fixed throughout.

For positive subinterval width, strong H^1 continuity follows from smooth dependence of short geodesics on endpoints and continuity of restriction and rescaling in L^2 . As the width tends to zero, the replacement energy is bounded by the original energy on the disappearing intervals. Those integrals tend uniformly to zero along any strongly H^1 -convergent sequence. This proves continuity also at $s = 0$. Centering the replacements proves the stated symmetries.

On the closed sublevel $f_N \leq B^2$, every edge has length at most $B/\sqrt{N} < \rho$, so this sublevel is compactly contained in P_N . Its boundary is smooth at a regular value. The polygon first variation imposes matching velocities at every vertex, so its critical points are exactly the smooth closed geodesics in the band.

At such a geodesic, decompose an H^1 variation field into its piecewise Jacobi interpolation and a field vanishing at every vertex. Integration by parts makes the two summands orthogonal for the index form. On a cell the second summand satisfies the Dirichlet Poincaré inequality. A curvature upper bound, the speed bound B , and sufficiently small cell width make its index form positive definite. The polygon Hessian is the first summand, which therefore has the full Hilbert index and nullity. Sard’s theorem for f_N gives dense regular cuts, since its critical values in the band are exactly those of E . Compact smooth manifolds with boundary have finite CW models, completing the claim. $\square$

If $k \mid N$, the fixed manifold of shift by $1/k$ in P_N is the repeated copy of $P_{N/k}$, and

$$f_N|_{P_{N/k}} = k^2 f_{N/k}. \quad (3.1)$$

In particular, the half-translation fixed manifold has energy $4f_{N/2}$. These identities follow directly from the sum defining f_N .

3.2 Prime images and iteration

Lemma 3.4 (Counting prime images). *On any smooth Riemannian manifold, every nonconstant closed geodesic is an iterate of a prime geodesic. Two prime geodesics with the same image differ by phase and possibly reversal. Reversal never preserves an oriented phase orbit.*

Proof. A nonconstant geodesic has constant positive speed and is an immersion. Its periods form a closed discrete subgroup of $\mathbb{R}$, so there is a least positive period. Dividing by that period gives its prime root.

To compare two primes with the same image, first use unit-speed parameters. Cover the parameter circle of the first by finitely many compact intervals on which it is an embedded arc. The inverse images of those arcs under the second geodesic form a finite closed cover of its parameter circle. By Baire’s theorem, one contains an interval. Within this interval choose a point whose image is not an endpoint of the chosen arc; an immersion cannot take an interval into that finite endpoint set. Near this point both geodesics lie in the same embedded arc and have the same tangent line. Their unit velocities agree up to sign. Uniqueness for the geodesic equation identifies them globally up to translation and reversal. Their least unit-speed periods agree, giving the assertion for period-one parametrizations as well. This proof allows self-intersections of either prime.

If reversal preserved an oriented phase orbit, one would have $\gamma(t) = \gamma(a-t)$. Differentiation at a fixed point of this parameter reflection gives $\dot{\gamma} = -\dot{\gamma}$, impossible for a nonconstant geodesic. $\square$

Suppose henceforth, when discussing a finite collection of critical labels, that there are only finitely many prime images. Let $c_1, \dots, c_p$ be prime representatives and $\ell_i > 0$ their lengths. By Lemma 3.4, the critical set at positive energy consists of the two oriented phase circles for each iterate c_i^m . Its critical F -values are $m\ell_i$. There are at most $2p$ oriented labels at a level and

$$O(1+b) \text{ labels below } b, \quad O(1+D) \text{ labels in an interval of width } D. \quad (3.2)$$

The constants depend only on the finite list of lengths. In particular, positive critical values are locally finite.

Write $I_{i,m}$ for the periodic Morse index of c_i^m . The following elementary form of the iteration estimate suffices; it is consistent with the sharper classical iteration theory [6].

Lemma 3.5 (Index and root types). *For each prime c_i , there is a number $\Delta_i \geq 0$ such that*

$$|I_{i,m+k} - I_{i,m} - I_{i,k}| \leq n, \quad |I_{i,m} - m\Delta_i| \leq n. \quad (3.3)$$

There is an integer $D_i \geq 1$ such that, for $s = \gcd(m, D_i)$, every null field at c_i^m repeats a null field at c_i^s . Thus only finitely many root kernels occur. For even m ,

$$|I_{i,m} - 2I_{i,m/2}| \leq n. \quad (3.4)$$

Proof. Use the periodic index form over integer intervals at prime speed. In the form domain for the $(m+k)$ -period problem, require that the value at time m equal the initial value. In the direct sum of the m - and k -period domains, require that the two initial values agree. Both are restrictions of codimension at most n , and the restricted forms identify. If the two indices drop by $a, b \in [0, n]$, their original difference is $a - b$. This proves the first estimate. The indices are finite since the index form is a positive derivative form plus a compact zeroth-order perturbation.

The sequences $I_{i,m} + n$ and $I_{i,m} - n$ are respectively subadditive and superadditive. Their normalized limits agree; applying the inequalities along multiples of each fixed m gives

$$\frac{I_{i,m} - n}{m} \leq \Delta_i \leq \frac{I_{i,m} + n}{m}.$$

Nonnegative indices give $\Delta_i \geq 0$. Equation (3.4) is the first estimate with two equal summands.

Let P_i be the real Jacobi Cauchy-data monodromy over one prime period. The periodic kernel is $\ker(P_i^m - 1)$. Take D_i to be the least common multiple of the orders of its root-of-unity eigenvalues, with value one if there are none. For $s = \gcd(m, D_i)$, the same eigenvalues satisfy $\lambda^m = 1$ and $\lambda^s = 1$. On a contributing complex Jordan block, $z^m - 1$ has a simple root, so its kernel is the ordinary eigenspace. All other blocks contribute zero. Hence

$$\ker(P_i^m - 1) = \ker(P_i^s - 1).$$

ODE uniqueness identifies the corresponding fields as repeated root fields. This computation is over $\mathbb{R}$ or $\mathbb{C}$; it makes no division by multiplicity in $\mathbb{F}_2$ -cohomology. $\square$

In particular, $\Delta_i = 0$ implies $0 \leq I_{i,m} \leq n$ for every m . We shall retain these iterates; they may occur at arbitrarily large lengths.

3.3 A common flow for local restriction maps

The next lemma is also useful for a compact Morse–Bott critical manifold. It does not require finitely many prime geodesics. Here a *good pair* means a closed pair whose second space is a strong deformation retract of an open neighborhood; its relative singular cohomology is the reduced cohomology of its pointed quotient.

Lemma 3.6 (Flow lenses). *Let a compact group H act on a finite-dimensional or Hilbert manifold, and let f be a smooth H -invariant function with a complete bounded H -equivariant strict descending flow ϕ . Suppose the critical set in a closed energy band is a finite disjoint union of compact sets at a single level c , and that H permutes these sets. Assume that the energy drop is bounded below away from every neighborhood of those sets, and that the corresponding assertion holds on the annuli of nested neighborhoods. These conditions hold for the normalized energy flow of Lemma 3.1, and on compact polygon bands.*

For a common sufficiently large T and sufficiently tight regular cuts $a < c < b$, set

$$\begin{aligned} X &= \{f \leq b\}, & X_- &= \{f \leq a\}, & f_T &= f \circ \phi_T, \\ Y &= \{x \in X : f_T(x) \leq a\}, & N &= \{x \in X : f_T(x) \geq a\}, & Q &= N \cap Y. \end{aligned} \quad (3.5)$$

There are natural relative cohomology isomorphisms, also after Borel construction, induced by

$$(X, X_-) \longrightarrow (X, Y) \longleftarrow (N, Q). \quad (3.6)$$

The lens splits into clopen pairs in the chosen disjoint critical neighborhoods. All maps commute with restriction to a fixed submanifold when the same flow is restricted there, with subgroup restriction, and with cup product by ambient absolute classes.

Proof. Choose disjoint neighborhoods W_α of the compact critical sets, and smaller V_α with positive distance from the complement of W_α ; the choices are equivariantly permuted. Work first in a fixed isolating band. Uniform drop outside the V_α makes a sufficiently long segment remaining in that band visit one of them. Traversing the annulus between V_α and the complement of W_α takes positive time, since the speed is bounded and its width is positive. The traversal costs a uniformly positive amount of energy. Choose the final width $b - a$ smaller than all these costs. Every segment of duration T with initial energy at most b and terminal energy at least a is then trapped in a single W_α , before and after its visit to V_α .

The first map in (3.6) is a pair homotopy equivalence: its inverse is the map induced by ϕ_T . The intermediate flow preserves the relevant subspaces. The finite closed cover $X = N \cup Y$ gives

$$N/Q \cong X/Y$$

by closed pasting. This does not require X to be compact. At $f_T = a$, the derivative of f_T along the flow is strictly negative. Transverse first hitting of this level is continuous, with hitting time zero on the lower side. Flowing for a fraction of that time gives a neighborhood retraction onto Y in X . On the upper side, stop at equality to obtain the corresponding retraction onto Q in N . The condition $f \leq b$ is preserved by descent. These are good pairs, so their quotient homeomorphism gives the second relative isomorphism. If a collapse set is empty, adjoin a disjoint basepoint.

The constructions are equivariant. Finite invariant closed covers remain closed covers after Borel construction, and the neighborhood homotopies descend. This proves the equivariant version without asserting compactness of a Borel space. Trapping makes $N \cap W_\alpha$ and $Q \cap W_\alpha$ clopen label pairs. Restriction to a fixed manifold intersects every space in (3.6) with that manifold, using the original flow. Finally, the maps in that diagram are inclusions and hence respect ambient cup products and connecting homomorphisms. $\square$

For the normalized energy flow, the drop hypotheses in Lemma 3.6 follow from Palais–Smale: a sequence away from the chosen critical neighborhoods with drop tending to zero would converge to a critical point outside them. For a finite polygon band they follow from compactness.

Theorem 3.7 (Uniform local control). *Assume there are finitely many prime images. There are constants $A, B < \infty$ such that every sufficiently tight crossing of a positive critical level has the following properties.*

- (i) *It decomposes into unoriented labels (i, m) at that level. Let $\mathcal{P}_{i,m}$ be the ordinary pair for one orientation. Then*

$$\sum_j \dim H^j(\mathcal{P}_{i,m}; \mathbb{F}_2) \leq B, \quad H^j(\mathcal{P}_{i,m}; \mathbb{F}_2) = 0 \quad \text{if } |j - I_{i,m}| > A.$$

- (ii) *Its reflection block is induced from $\mathcal{P}_{i,m}$; its K -block is induced from the $\langle \tau \rangle$ -action on that pair. For odd m the fixed label is empty. For even m , the ordinary fixed pair has total rank at most B and support $|j - I_{i,m/2}| \leq A$.*
- (iii) *Full-to-fixed restriction respects the labels. These identifications are compatible with ambient maps and connecting homomorphisms.*
- (iv) *Cup product by $e = \text{ev}_0^*[S^n]^* \in H_R^n(\Lambda; \mathbb{F}_2)$ vanishes on the reflection crossing, including each individual label.*

The constants depend on the metric and the finite prime list, and include all degenerate iterates and all zero-mean-index primes.

Proof of the decomposition, naturality, and cup assertion. Choose an even polygon order adequate for the event. Start with one complete K -invariant strict descending gradient flow for the original f_N . Its gradient is tangent to $\text{Fix}(\tau)$. Thus a critical point of the restriction is exactly a full critical point there, and the restricted flow is strict away from them.

By Lemma 3.4, the critical circles come in disjoint reversed pairs. The half-shift preserves each orientation circle and fixes it pointwise exactly when m is even; for odd m , choose its neighborhood disjoint from the fixed locus. Use Lemma 3.6 with disjoint equivariantly permuted neighborhoods of these oriented circles. Reflection induction identifies a block with the ordinary pair of one orientation, while K induction retains its half-shift action. Literal intersection of the original lenses with the fixed manifold gives the full-to-fixed map. In particular, the actual diagrams are

$$\begin{array}{ccc} H_K^j(X, X_-) & \xrightarrow{\cong} & H_K^j(N, Q) \\ \text{res} \downarrow & & \downarrow \text{res} \\ H_K^j(X^\tau, X_-^\tau) & \xrightarrow{\cong} & H_K^j(N^\tau, Q^\tau), \end{array}$$

with $N^\tau = N \cap \text{Fix}(\tau)$ and likewise for Q . They commute with ambient relative maps and the associated exact sequences. No alternative local normal form is being used to identify these maps.

The finite union of prime images does not fill S^n , for $n \geq 2$. Choose q outside it, and choose all critical neighborhoods inside the invariant open set

$$\bigcap_{g \in K} g^{-1}(\text{ev}_0^{-1}(S^n \setminus \{q\})).$$

The entire lens therefore evaluates into the punctured sphere. Evaluation at zero is reflection invariant, so it induces $ER \times_R N \rightarrow S^n \setminus \{q\}$. The top sphere class pulls back to zero. Hence its cup product on $H_R^*(N, Q)$ is zero. The inclusion isomorphisms of Lemma 3.6 transfer this vanishing to the actual crossing. This uses only reflection invariance of evaluation; half-translation invariance is not needed. All choices are common to the finitely many labels at a simultaneous level. The uniform rank statements are proved next. $\square$

3.4 Computing ranks while retaining the original maps

The finite-root reduction is the local mechanism in classical iteration arguments [16]. We give the required rank calculation explicitly, including its relative Thom pair.

Fix one oriented label at energy c . Add small nonnegative bumps, constant near the other circles at c , supported away from the selected neighborhood W . On their compact transition regions the original strict drop is bounded away from zero. Taking the amplitudes sufficiently small leaves the same original flow strict for the modified energy f_i . No new critical points appear, and only the selected circle remains in a sufficiently narrow band about c .

Choose one large time and tight cuts for the original energy and this finite list of bumped energies. The total lens for f_i is exactly its original labeled lens: in either direction every relevant segment is trapped in W , where the functions agree. The same construction on the fixed manifold identifies its ordinary label with an isolated crossing. Its ordinary rank is now independent of the strict flow and of smaller regular cuts, by strip deformations. We may compute these ranks using a split flow. The original flow remains the one defining all maps in Theorem 3.7.

Lemma 3.8 (The normal shift). *Put $s = \gcd(m, D_i)$ and $k = m/s$. The ordinary local rank at c_i^m , in degree j , equals the ordinary local rank of the corresponding root label at c_i^s , in degree $j - (I_{i,m} - I_{i,s})$.*

Proof. Normal coordinates. Take N divisible by k and sufficiently large. The fixed manifold $Z = P_{N/k} \subset P_N$ of the $1/k$ -shift has restricted energy $h = k^2 f_{N/k}$. Near the selected orbit the bump vanishes. Use a cyclically invariant tubular metric. Symmetry makes the normal first derivative vanish along Z . The Hessian has reducing fixed and normal summands, and Lemma 3.5 places its full kernel inside the fixed tangent space. The normal block is therefore nonsingular, of negative rank

$$b_- = I_{i,m} - I_{i,s} \geq 0.$$

We record the normal coordinate construction to keep the degenerate base unchanged. Taylor integration writes

$$f_i(z, \nu) = h(z) + \langle A(z, \nu)\nu, \nu \rangle,$$

where $A_0(z) = A(z, 0)$ is symmetric and invertible. Normalize by $|A_0|^{1/2}$, putting

$$B = |A_0|^{-1/2} A |A_0|^{-1/2}, \quad J = \text{sign}(A_0).$$

Near the zero section, B is close to J . The near-identity series $D = (JB)^{1/2}$ is J -self-adjoint, and $D^*JD = B$. Thus the fiber inverse function theorem applied to $D|A_0|^{1/2}\nu$ gives

$$f_i = h(z) - |u|^2 + |y|^2. \tag{3.7}$$

The construction is uniform near the compact orbit. The positive and negative bundles can be nontrivial; no trivialization of them is required.

The local quotient. Let ψ_t be a complete strict descending flow for the unmodified h . In (3.7), lift it using metric connections, expand u exponentially, and contract y exponentially. The energy derivative is

$$dh(\dot{\psi}) - 2|u|^2 - 2|y|^2 < 0$$

off the critical set. Convex interpolation with a normalized negative gradient gives a complete strict full field agreeing with this one near the orbit.

Choose a full trap W_f whose base projection has compact closure inside the base chart and avoids all other h -critical points. The exact-field region contains uniform fiber balls over this closure. A smaller selected base trap W_h lies under uniform small fiber balls contained in W_f ; all other base traps lie outside its projection. First choose a common sufficiently large time T for full and base trapping, then sufficiently tight regular cuts $a < c < b$. In particular, $\sqrt{b-a}$ is smaller than the available fiber radius.

Apply Lemma 3.6 to the isolated full crossing. On its quotient $N/Q = X/Y$, shrink y to zero. The new first T -paths have the same base and negative coordinate, and a scaled positive coordinate, so they stay in the exact region. Both initial and terminal energies decrease. Thus the maps take N into X and Q into Y , and descend continuously to the quotient; some points may collapse. This retracts onto the $y = 0$ subset together with the basepoint.

Apply the full forward flow to this subset. While a point remains uncollapsed, every shifted T -segment is trapped in the exact region, so $y = 0$ persists and the base follows ψ . On the compact base projection where $h \geq b > c$, its drop has a positive lower bound. A single further time therefore sends all survivors into $h \leq b$. This last subset is preserved up to collapse, so its inclusion is a homotopy equivalence. The homotopies are continuous on the pointed quotient; products of a quotient map with the compact interval remain quotient maps.

Write $h_T = h \circ \psi_T$. The invariant subset $h \leq b$ used in this homotopy equivalence, with its equality boundary before collapse, is described by

$$P = \{h \leq b, h_T \geq a\}_{\text{selected}}, \quad A_P = \{h_T = a\} \cap P, \quad |u| \leq e^{-T} \sqrt{h_T(z) - a}. \quad (3.8)$$

For one direction, the projected segment stays in the base band and the projection of the full trap, so it belongs to the selected base label. Conversely, a selected base path is trapped in W_h , and $e^t|u| \leq \sqrt{b-a}$ for $0 \leq t \leq T$. Its lift therefore stays in the exact region and satisfies the full lens inequalities. Equality in the radius bound is precisely its exit.

Let D_- and S_- be the unit disk and sphere bundles of the negative bundle over P . Scaling by the continuous radius in (3.8) gives a homeomorphism from

$$D_- / (S_- \cup (D_-)_{A_P}) \quad (3.9)$$

onto the final pointed subset. Indeed it is bijective away from the collapsed union; its domain is compact and its target is Hausdorff. The radius vanishes exactly over A_P . This also handles rank zero and the disjoint-basepoint convention.

Relative Thom calculation. For completeness, we prove the relative Thom calculation for this lens, without requiring a manifold structure on its base. On P , let t_p be the first forward time until $h_T \leq a$, truncated at a fixed $T_0 > 0$. It is continuous. Finite hits are transverse since a is regular; at a point that has not hit by T_0 , either the hit occurs transversely at T_0 , or the strict terminal inequality persists in a neighborhood. The base flow stays in its clopen label until equality.

For a unit-disk coordinate put

$$t(z, u) = \min\{t_p(z), -\log|u|\}, \quad -\log 0 = +\infty.$$

This is continuous also at $u = 0$, where the radial entry is inactive for $|u| < e^{-T_0}$. Its zero set is $\mathcal{A} = S_- \cup (D_-)_{A_P}$. On the open neighborhood $t < T_0$, flow the base for time λt , parallel transport u , and multiply its norm by $e^{\lambda t}$. Neither the base exit nor the sphere is passed before the stopping time. The residual minimum time is $(1 - \lambda)t$, so the neighborhood is

preserved, the union is fixed, and at $\lambda = 1$ it is reached. This proves that $(D_-, \mathcal{A})$ is a good pair. Rank zero simply removes the radial alternative.

The sphere pairs are good by radial collars. Closed pasting identifies

$$H^*(\mathcal{A}, S_-) \cong H^*((D_-)_{A_P}, (S_-)_{A_P}).$$

Under the natural absolute mod-2 Thom isomorphisms, restriction from $H^*(D_-, S_-)$ to this group is the restriction $H^{*-b-}(P) \rightarrow H^{*-b-}(A_P)$. The exact sequence of the triple $S_- \subset \mathcal{A} \subset D_-$ gives

$$\begin{aligned} 0 &\longrightarrow \text{coker}(H^{j-b-1}(P) \rightarrow H^{j-b-1}(A_P)) \longrightarrow H^j(D_-, \mathcal{A}) \\ &\longrightarrow \ker(H^{j-b-}(P) \rightarrow H^{j-b-}(A_P)) \longrightarrow 0. \end{aligned}$$

The base-pair exact sequence has the same kernel and cokernel. Thus

$$\dim H^j(D_-, \mathcal{A}) = \dim H^{j-b-}(P, A_P). \quad (3.10)$$

This argument compares vector spaces, rather than subtracting possibly infinite dimensions.

Absolute Thom applies to these compact metrizable bases as follows. Pull the bundle back along the CW weak equivalence $|\text{Sing } P| \rightarrow P$. The disk and sphere bundle projections are fibrations, so their induced total-space maps are weak equivalences. They give singular cohomology isomorphisms, including the relative disk/sphere groups and their restriction maps. Apply the usual mod-2 Thom isomorphism on the CW model, naturally under the same construction for A_P . The rank-zero case is the base pair itself.

Finally, divide the base cuts by k^2 in (3.1). The selected base label is the root label at c_i^s , so (3.10) proves the asserted normal shift. Iteration is a homeomorphism onto the repeated-loop subspace, including its oriented phase circles; this identification uses no covering transfer or division by multiplicity. $\square$

Completion of Theorem 3.7. There are only finitely many root levels $(s\ell_i)^2$, with $s \mid D_i$. Around each distinct level choose a fixed isolated interval with regular endpoints. Choose one coarse polygon order N_0 adequate up to every upper endpoint.

The total crossing pairs in P_{N_0} have finite total cohomological rank: their regular sublevels are compact smooth manifolds with boundary, and the pair exact sequence applies. Their relative cohomology vanishes outside $0, \dots, nN_0 + 1$. Every selected root label is a direct summand of such a total crossing by Lemma 3.6. This includes coincident root levels.

At an arbitrary event, choose the fine polygon order divisible by kN_0 for every label, where $k = m/\gcd(m, D_i)$. For every even label also impose divisibility by $2k'N_0$, where

$$k' = \frac{m/2}{\gcd(m/2, D_i)}.$$

These are finitely many conditions. After the normal shift, polygon retractions through the Hilbert pair identify each finer root crossing with its coarse pair. Tight cuts within the fixed isolating intervals give the same groups by regular-strip deformations. The finite maximum of their total ranks is therefore a bound B for every iterate label.

If the root group in degree a is nonzero in the shift calculation, then

$$j - I_{i,m} = a - I_{i,s}, \quad 0 \leq a \leq nN_0 + 1.$$

For example,

$$A = nN_0 + 1 + \max_{i, s \mid D_i} I_{i,s}$$

is a uniform support constant. Enlarge the constants if necessary for the fixed computations: on $\text{Fix}(\tau)$ start with $4f_{N/2}$, apply the same argument to $m/2$, and divide cuts by $4(k')^2$. The support is centered at $I_{i,m/2}$.

The normal shift changes no total rank. All degeneracy remains in finitely many root crossings, so the proof assumes neither nondegeneracy nor analyticity. It never divides by Δ_i ; zero-mean-index primes satisfy the same conclusions. $\square$

The uniform total-rank assertion is for ordinary local groups. The half-shift Borel group may have an infinite polynomial tail. The Gysin sequence of Proposition 7.1 converts the ordinary bound into a uniform bound in each equivariant degree and an eventual multiplication isomorphism. The finite-cut comparison will use these consequences together with the original, label-preserving restriction maps established here.

4 Shortening families in a fixed degree

We first prove a filling estimate whose constant may depend on the homological degree. The geometric argument works on every compact simply connected Riemannian manifold. Its two ingredients are continuous arclength parametrization and a shortening procedure with only finitely many simultaneous changes. Throughout this section, chains have coefficients in $\mathbb{F}_2$; for the manifold M under consideration, write $\Lambda = H^1(S^1, M)$ and $\mathcal{A}_a = \{\gamma \in \Lambda : \ell(\gamma) < a\}$. Block, Manin and Weinberger established fixed-degree additive filling bounds for based and free Lipschitz loop spaces [5, Theorem 5.1]. We give a direct proof in the strong H^1 topology for the length filtration used here.

4.1 Arclength parametrization in H^1

Lemma 4.1 (Continuous arclength homotopy). *Let M be a compact Riemannian manifold. There is a jointly continuous map*

$$\Psi : [0, 1] \times H^1([0, 1], M) \longrightarrow H^1([0, 1], M)$$

in the strong H^1 topology such that Ψ_0 is the identity, $\Psi_1\gamma$ has constant speed, and every Ψ_u preserves length and endpoints. Constant paths and constant-speed paths are fixed. Moreover,

$$E(\Psi_u\gamma) \leq (1 - u)E(\gamma) + u\ell(\gamma)^2 \leq E(\gamma). \quad (4.1)$$

The homotopy commutes with reversal. In particular, on loops it preserves evaluation at zero and the constant-loop subspace.

Proof. Write $v(t) = |\dot{\gamma}(t)|$ and $L = \int_0^1 v(t) \, dt$. When $L > 0$, put

$$\sigma(t) = \frac{1}{L} \int_0^t v(s) \, ds, \quad \phi_u(t) = (1 - u)t + u\sigma(t).$$

For $u < 1$, the absolutely continuous map ϕ_u is an increasing homeomorphism with Lipschitz inverse. Set $\Psi_u\gamma = \gamma \circ \phi_u^{-1}$. At $u = 1$ the inverse need not exist. Instead, define $\Psi_1\gamma(s)$ to be the value of γ on the fiber $\sigma^{-1}(s)$. Each fiber is an interval on which $v = 0$ almost everywhere, so absolute continuity makes this definition unambiguous. For $L = 0$ leave the constant path fixed for every u .

For $s_1 < s_2$, choose ordered preimages $t_1 < t_2$ under σ . Then

$$d(\Psi_1\gamma(s_1), \Psi_1\gamma(s_2)) \leq \int_{t_1}^{t_2} v(t) \, dt = L(s_2 - s_1).$$

Thus $\Psi_1\gamma$ is L -Lipschitz. Since $\gamma = (\Psi_1\gamma) \circ \sigma$ and σ is a nondecreasing surjection, comparison of partitions gives $\ell(\gamma) \leq \ell(\Psi_1\gamma)$. The Lipschitz estimate gives the reverse inequality. Consequently $\Psi_1\gamma$ has length L and speed L almost everywhere, and its energy is L^2 .

For $u < 1$, change of variables by ϕ_u gives

$$E(\Psi_u\gamma) = \int_0^1 \frac{v(t)^2}{(1-u) + uv(t)/L} dt.$$

On $\{v > 0\}$, convexity of the reciprocal bounds the integrand by $(1-u)v^2 + uLv$; on $\{v = 0\}$ it is zero. This proves (4.1), including $u = 1$ by the preceding argument. The same change of variables preserves length for $u < 1$. Endpoints are unchanged, and constant-speed paths have $\sigma(t) = t$.

We verify joint strong continuity, including the possibly flat fibers at $u = 1$. Fix a smooth embedding of M in a Euclidean space and extend its metric to a smooth positive definite matrix on a neighborhood of M . Suppose $\gamma_i \rightarrow \gamma$ strongly in H^1 and $u_i \rightarrow u$. The one-dimensional Sobolev embedding gives uniform convergence of the paths. Smoothness of the metric gives $v_i \rightarrow v$ in L^2 , and hence $L_i \rightarrow L$.

If $L > 0$, the maps σ_i and ϕ_{u_i} converge uniformly to σ and ϕ_u . Given $s_i \rightarrow s$, choose t_i in the inverse fiber defining $\Psi_{u_i}\gamma_i(s_i)$. Every subsequential limit t satisfies $\phi_u(t) = s$. The path γ has one value on that fiber: for $u < 1$ the fiber is a singleton, and for $u = 1$ this was proved above. Therefore

$$\Psi_{u_i}\gamma_i(s_i) = \gamma_i(t_i) \rightarrow \Psi_u\gamma(s).$$

A compactness argument makes this convergence uniform in s . If $L = 0$, the image of each output is contained in the image of its input, so uniform convergence to the constant path follows directly.

The energies also converge. If $L > 0$ and $u < 1$, the reciprocals in the integral formula are uniformly bounded and converge in measure; combining this with $v_i^2 \rightarrow v^2$ in L^1 proves convergence of the integrals. If $u = 1$, Cauchy–Schwarz and (4.1) give

$$L_i^2 \leq E(\Psi_{u_i}\gamma_i) \leq (1-u_i)E(\gamma_i) + u_iL_i^2 \rightarrow L^2.$$

If $L = 0$, the upper bound $E(\Psi_{u_i}\gamma_i) \leq E(\gamma_i)$ tends to zero.

For completeness, energy convergence here implies strong, rather than merely weak, convergence of derivatives. Uniform convergence and the energy bound give weak H^1 convergence to the already identified output. In the squared derivative norms, replace the metric matrix at the varying outputs by its value at the limiting output. The resulting error tends to zero, because the matrices converge uniformly and the derivatives are bounded in L^2 . The derivative norms thus converge in one fixed positive weighted L^2 Hilbert norm. Weak convergence together with convergence of these norms gives strong convergence. This proves the asserted joint continuity.

Finally, for the reversed path the arclength map is $1 - \sigma(1 - t)$. This identity proves commutation with reversal, also at the quotient endpoint $u = 1$. $\square$

We shall use this homotopy only for ordinary homology. Fixed-time concatenation of finitely many H^1 paths is continuous and adds their lengths; Lemma 4.1 then puts the concatenation at constant speed. The same is true when some paths are constant. Restriction to a moving initial interval, followed by linear rescaling, is also continuous in H^1 , including a collapsed interval: for $\gamma_t(s) = \gamma(ts)$,

$$E(\gamma_t) = t \int_0^t |\dot{\gamma}(r)|^2 dr \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

Continuity away from zero follows from continuity of dilation in L^2 .

4.2 A bounded local path replacement

The following elementary construction is the source of the uniform constant in the shortening argument.

Lemma 4.2 (Local replacement of paths). *Fix $L_0 < \infty$ and let M be compact and simply connected. Pairs of paths with common endpoints and Lipschitz constants at most L_0 admit a finite cover by relative C^0 neighborhoods in this path-pair space with the following property. On each neighborhood there is a relative-endpoint homotopy from the first path to the second, continuous in the strong H^1 topology of the input paths and the homotopy parameter. Every path occurring in these homotopies has length at most a constant $K = K(M, L_0)$.*

Proof. The space of such pairs is compact in C^0 by the Arzelà–Ascoli theorem. Consider one pair (α_0, β_0) from this space, with endpoints x_0, y_0 . Simple connectivity gives a continuous homotopy between them, fixing the endpoints. Replace its paths by polygons on one sufficiently fine, fixed subdivision of the path parameter. Uniform continuity of the original homotopy ensures that all consecutive vertices are within the injectivity radius. Polygonal interpolation then gives an H^1 -continuous homotopy with a finite length bound. Short pointwise geodesic interpolation joins its two endpoint polygons to α_0, β_0 . These interpolations also have a finite length bound, since the original endpoint paths are Lipschitz.

This construction extends to nearby pairs with varying common endpoints x, y . Attach the short geodesics from x to x_0 and from y_0 to y to each prototype path, using fixed, small positive time intervals at its ends. First choose those intervals small enough that the retimed prototype endpoints are uniformly close to the original endpoints; this uses their Lipschitz bounds. Then restrict x, y and the input paths to a sufficiently small C^0 neighborhood. Short pointwise geodesic interpolation joins the actual first path to the adjusted prototype first path, and similarly at the other end. Both interpolations fix x, y . Their derivatives are bounded by a fixed multiple of the derivatives of the two interpolated paths. Consequently their lengths have a common bound on this neighborhood. Every operation just described is continuous in the strong H^1 topology. A finite subcover of the compact pair space, and the maximum of its finitely many bounds, give K . $\square$

Here and below a local homotopy is a specified choice on a neighborhood; no global choice of a minimizing geodesic is needed. We next use these local choices to shorten a long path block with bounded overhead.

Lemma 4.3 (Shortening one block). *Let M be compact and simply connected. There are constants $C, D > 0$, depending only on M , with the following property. Suppose a path is divided into finitely many pieces, each of which, on a unit interval, has Lipschitz constant at most one. Near any such path, within this space of piecewise Lipschitz paths, there is a relative-endpoint homotopy to a path of length at most D . At every stage its length is at most its original length plus C . The homotopy is continuous in H^1 on that neighborhood. The constants are independent of the number of pieces.*

Proof. Set $D = \text{diam}(M) + 1$. For a fixed prototype choose shortest paths, called spokes, from its initial point to all its piece vertices. For nearby inputs extend these choices by short endpoint connectors, using fixed thirds of the unit interval. Shrink the neighborhood so that each spoke has length at most D and Lipschitz constant at most $3D$. At the initial vertex use the constant spoke.

Proceed through the pieces in order. Assume the preceding prefix has already been replaced by its spoke. Retime that spoke and the next piece to halves of the enlarged prefix. This retiming preserves length and is continuous in H^1 : every nonconstant path has a positive time interval throughout the retiming. In the first step, where the prefix is empty, insert a

constant wait at the initial point and retime the first piece. The two paths to be compared are now the spoke followed by the next piece, and the spoke to the next vertex. On a unit interval both have Lipschitz constant at most

$$L_0 = 2 \max\{1, 3D\}.$$

Apply Lemma 4.2 to replace the first by the second, and leave the remaining suffix unchanged.

During a replacement the active prefix has length at most $K(M, L_0)$; during a re-timing it has length at most $D + 1$. The suffix still consists of original pieces. Thus $C = \max\{K(M, L_0), D + 1\}$ bounds the increase over the length of the entire original path at every stage. At the end the path is its final spoke, of length at most D . For a particular input there are finitely many pieces, hence finitely many local choices. Intersecting their neighborhoods makes the entire construction continuous on one neighborhood. This affects the neighborhood, but neither C nor D . $\square$

The intermediate energies in Lemma 4.3 need not have a uniform bound: a long prefix may be assigned a short time interval. Only its length estimate is used. All time intervals for nonconstant pieces remain positive, and the construction is an H^1 homotopy. Lemma 4.1 restores constant speed whenever required.

4.3 Coloring a finite family

Lemma 4.4 (A fixed-dimensional family shortens by one). *Let M be compact and simply connected. For each integer $j \geq 0$ there are $a_j, H_j > 0$ such that every H^1 -continuous family $\Gamma : P \rightarrow \mathcal{A}_a$, where P is a finite polyhedron of dimension at most j and $a \geq a_j$, is homotopic to a family in $\mathcal{A}_{a-1}$ through $\mathcal{A}_{a+H_j}$. The constants do not depend on P or on the number of its simplices.*

Proof. Normalize the family by Lemma 4.1. A loop of length $L < a$ now has constant speed L . Let C, D be the constants of Lemma 4.3. Choose B so large that

$$B/2 > D + jC + 2. \tag{4.2}$$

For sufficiently large a , the integer $k = \lfloor a/B \rfloor$ is at least $j + 1$. Divide the loop into k equal time blocks, and each block into $\lceil a/k \rceil$ equal pieces. Every piece, rescaled to a unit interval, has Lipschitz constant at most one. Every block has length L/k .

For each parameter point, Lemma 4.3 supplies local recipes for the first $j + 1$ blocks. There are only finitely many blocks and pieces in this particular family. Choose a common neighborhood for these recipes, and cover P by finitely many such neighborhoods. Subdivide a triangulation finely enough that the closed stars in a final barycentric subdivision are subordinate to this cover. Color each vertex by the dimension of the simplex of the preceding triangulation whose barycenter it is. The colors lie in $\{0, \dots, j\}$ and are distinct on each simplex.

At a vertex v of color h , use its local recipe on block $h + 1$. If λ_v is its barycentric coordinate, at homotopy time s run that recipe for the fraction

$$s \min\{1, (j + 1)\lambda_v\}.$$

Two positive-coordinate stars of the same color are disjoint; at the boundary of either star the fraction is zero and the recipe is the identity. Recipes of different colors affect different blocks, keeping their endpoints fixed. They therefore paste to one continuous H^1 homotopy. At each parameter at most $j + 1$ blocks are changed, and the length throughout is less than $a + (j + 1)C$.

At the terminal time some barycentric coordinate is at least $1/(j+1)$, so one block has been completely shortened. The final length is consequently at most

$$L - L/k + D + jC. \quad (4.3)$$

If $L \geq a/2$, then $k \leq a/B$ gives $L/k \geq B/2$, and (4.2) makes (4.3) less than $a - 1$. If $L < a/2$, the rough bound $L + (j+1)C < a/2 + (j+1)C$ has the same consequence once $a > 2((j+1)C + 1)$. Choose a_j larger than this threshold and large enough that $k \geq j+1$. We may take $H_j = (j+1)C$. $\square$

4.4 Filling the remaining bounded family

Lemma 4.5 (A fixed bounded level). *For every $b > 0$ and $j \geq 0$ there is $R_{b,j} < \infty$ such that every finite j -cycle in $\mathcal{A}_b$ which is null-homologous in Λ bounds a finite chain in $\mathcal{A}_{R_{b,j}}$.*

Proof. Normalize by Lemma 4.1; the resulting cycle has $F = \ell < b$. We describe a single finite model for all such normalized cycles. Choose an integer N large enough that paths of energy at most $(b+1)^2$ move less than a fixed small injectivity radius on intervals of length $1/N$. Sampling at these times and joining successive vertices by minimizing geodesics gives a continuous polygonal map. For the discrete energy

$$E_N(x_0, \dots, x_{N-1}) = N \sum_{i=0}^{N-1} d(x_i, x_{i+1})^2, \quad x_N = x_0,$$

Cauchy–Schwarz gives $E_N \leq E$ on sampled loops. Choose $b' \in (b, b+1)$ such that $(b')^2$ is a regular value of E_N . The set

$$Q = \{E_N \leq (b')^2\}$$

is contained strictly inside the open set of vertex tuples whose successive distances are smaller than the injectivity radius. It is a compact smooth manifold with boundary and therefore has finite-dimensional homology in every degree. Polygonal interpolation maps Q to loops of length at most b' .

There is also a homotopy from each normalized loop to its sampled polygon, with one uniform energy bound depending only on b and M . Indeed, on each interval both paths stay in a small normal neighborhood of its initial vertex. Pointwise short-geodesic interpolation is H^1 -continuous, and the derivative of this interpolation is bounded by a fixed multiple of the two path derivatives. Its square-root energy is thus at most $2C_0b$ for a fixed metric constant C_0 . This also gives a uniform length bound for the homotopy.

The kernel of $H_j(Q; \mathbb{F}_2) \rightarrow H_j(\Lambda; \mathbb{F}_2)$ is finite-dimensional. Choose finite cycles representing a basis and finite global fillings of their polygonal images. Their supports are compact, so their lengths have a common finite bound. The sampled image of any cycle in the lemma represents a linear combination of this basis. The difference bounds in Q , whose entire image has length at most b' . Adding that comparison chain, the finitely many chosen fillings, and the two normalization and polygonal homotopies proves the claim with one $R_{b,j}$ strictly larger than all these bounds. No bound on the number of simplices in the original cycle is needed. $\square$

Proposition 4.6 (Fixed-degree filling). *Let M be compact and simply connected. For every integer $j \geq 0$ there is $D_j > 0$ such that, for every real $a > 0$, each finite j -cycle in $\mathcal{A}_a$ which is null-homologous in Λ bounds a finite chain in $\mathcal{A}_{a+D_j}$.*

Proof. A finite singular cycle over $\mathbb{F}_2$ can be regarded as the image of a cycle on a finite polyhedron of dimension at most j . One explicit construction takes the finitely many singular

simplices, pairs equal faces that cancel in their boundary, and glues these faces by their ordered affine identifications. The first barycentric subdivision has embedded closed simplices and vertices distinguished by their original face dimensions; a second subdivision removes multiple simplices with the same vertex set. This gives a finite polyhedron with its top-dimensional cycle and the original map to Λ . The usual subdivision and prism chains identify its image with the original singular cycle. All these chains have image in the union of the original simplices. Thus the number of parameter simplices plays no role in a length estimate.

Apply Lemma 4.4 repeatedly, normalizing between successive applications. Starting at level a , the successive starting levels are $a, a-1, a-2, \dots$, until the family lies below one fixed level $b_j > a_j + 1$. The prism chains at the successive stages all lie in $\mathcal{A}_{a+H_j}$: the peak at stage r is bounded by $a-r+H_j$, so these bounds are not added over the stages. Only finitely many stages are needed for each input cycle.

The resulting cycle is still globally null-homologous. Apply Lemma 4.5 at the fixed level b_j . Taking $D_j > \max\{H_j, R_{b_j,j}\}$ gives the required filling after adjoining the shortening prisms. If the original a is already at most b_j , the bounded-level lemma applies immediately. This proves the assertion for every positive level, without a regular-value assumption. $\square$

The constants D_j presently depend on the degree. For spheres, the cut-join operations in the next section reduce all sufficiently large degrees to a fixed finite range. This will turn Proposition 4.6 into one filling bound valid in every degree.

5 A filtered cut-join identity

We now construct a degree-lowering operation which removes a long part of a loop. Its essential property is an interchange identity with the loop product, with an additive length cost independent of the degree. Applied to the sphere's product powers, this identity makes the cut of a suitable cycle equal to the constant-loop fundamental class. These constructions apply to every smooth metric on S^n , $n \geq 3$; no assumption about the number of closed geodesics is made in this section.

We retain $\mathcal{A}_a = \{\ell < a\}$ and $d = n-1$. A chain is always a finite singular chain with coefficients in $\mathbb{F}_2$. Writing $c \simeq c'$ in a subspace means that their difference bounds a chain in that subspace.

5.1 The sphere classes and their length thresholds

Write $\bullet$ for the Chas-Sullivan homology product [8], of degree $-n$, and $\otimes$ for the Goresky-Hingston product on $H^*(\Lambda, M)$ [15], of degree $n-1$. We use the following precise consequences of sphere resonance and the calculation of these products.

Proposition 5.1 (Sphere powers and resonance). *There are constants $\alpha > 0$ and C_H depending on g such that*

$$|k - \alpha \operatorname{cr}(z)| \leq C_H \quad (5.1)$$

for every nonzero homogeneous ordinary homology or cohomology class z of degree k . For homology, cr is the infimum of the levels a carrying the class in $X_a = \{F < a\}$; for cohomology it is the supremum of the levels on which its restriction vanishes. Moreover, each ordinary homology and cohomology group in a fixed degree has dimension at most one. There are classes

$$\Theta \in H_{3n-2}(\Lambda), \quad \omega \in H^{n-1}(\Lambda, M), \quad \theta \in H^{3n-2}(\Lambda, M)$$

with $\langle \theta, \Theta \rangle = 1$ such that, for every $m \geq 1$,

$$|\Theta^{\bullet m}| = n + 2md, \quad |\omega^{\otimes m}| = (2m - 1)d, \quad (5.2)$$

$$\langle \theta \otimes \omega^{\otimes m}, \Theta^{\bullet(m+1)} \rangle = 1. \quad (5.3)$$

In particular these powers are nonzero, and $H_n(\Lambda) = \mathbb{F}_2[M]$.

Proof. The resonance statement is Theorem 1.1 of [19]. Its cohomological critical levels may be computed in singular cohomology: the open energy sublevels and Λ are paracompact, locally contractible Hilbert manifolds, where singular and Čech cohomology agree. Their Lemma 5.3 gives the rank bound; Lemma 5.4 and Equation (12) give the powers and the pairing, with the index in that equation replaced by $m + 1$. These statements apply for every $n > 2$ and every field. The indicated powers and pairing may also be taken integrally and reduced modulo two: the integral pairing is one, so its reduction is one in both parities of n . The absolute class of degree $3n - 2$ has a unique relative lift because $3n - 2 > n + 1$. Finally evaluation retracts the inclusion of constant loops, so $[M] \neq 0$; the rank bound gives the last assertion. $\square$

For sufficiently large m , set

$$q = 2md, \quad p = q - d, \quad l = \frac{n + q + C_H}{\alpha} + 2, \quad L = \frac{p - C_H}{\alpha} - 2 > 0. \quad (5.4)$$

We also require $p > n + 1$. Choose a cycle S representing $\Theta^{\bullet m}$ in $\mathcal{A}_l$. Such a cycle exists already in X_l by (5.1). Choose a normalized cocycle T representing $\omega^{\otimes m}$ and vanishing on $\mathcal{A}_L$. Indeed its absolute class, identified with the relative class since $p > n + 1$, vanishes on X_L by resonance. The cohomology exact sequence gives a cocycle relative to X_L . Pull it back by the arclength map of Lemma 4.1; this sends $\mathcal{A}_L$ into X_L , fixes constants, and is homotopic to the identity. Normalization preserves the relative condition. Notice the fixed difference

$$l - L = \frac{2n - 1 + 2C_H}{\alpha} + 4, \quad (5.5)$$

independent of m . This is the length available for the eventual degree reduction.

5.2 Supported caps on small chains

We use unnormalized singular chains and normalized cochains. The shuffle cross product is associative and, over $\mathbb{F}_2$, symmetric under permutation of factors. Our cap evaluates a cochain on the front face and retains the back face. Thus successive caps correspond to cup products in the same order, and

$$\partial(c \frown b) = (\partial c) \frown b + c \frown \delta b.$$

The degenerate chains form an acyclic subcomplex in every subspace; the usual normalization homotopies preserve simplex images. Consequently an equality modulo degenerate chains between cycles gives a homology in the same length sublevel.

All tubes, arcs and rotations in the construction use a fixed round metric on M ; all length estimates use g . Put

$$F_\eta = \{(z, z') \in M^2 : d_{\text{rd}}(z, z') \geq \eta\}.$$

A cochain relative to F_η vanishes on every simplex contained in this set. Its nonzero evaluation on a face therefore supplies a point of that face with distance less than η . Subdividing a finite parameter family sufficiently finely makes the corresponding distance less than 2η on the entire parent simplex. For product chains we impose this condition on the whole product

rectangles before shuffling. Maps used below are defined on the larger 4η tubes, so uniform continuity on an intermediate closed tube allows successive support conditions to be imposed together.

The following observation also deals with cochains whose maps are defined only on a tube.

Lemma 5.2 (Supported cochain calculus). *Let U be a pulled-back Thom cocycle, and let h be a continuous map from its larger tube O to a space carrying a cocycle T . Extend h^*T by zero on simplices not wholly contained in O , and write $b = U \smile \widetilde{h^*T}$. Subdivide the finite parameter families factorwise so that a nonzero U -face forces its entire product rectangle into a fixed closed intermediate tube contained in O . Its preimage in each finite compact parameter domain is compact. Then $\delta b = 0$ on the shuffled simplices and every later face or image-preserving comparison prism in these rectangles.*

If T vanishes on loops of length less than L , the subdivision may also ensure the following: a contributing term with a nonzero U -face and a nonzero T -face has $\ell(h) \geq L - \varepsilon$ throughout its whole rectangle, for any fixed $\varepsilon > 0$. The same conclusion holds for the block and skip faces used by cup-one, and for all later image-preserving subdivision prisms.

Proof. Extension by zero defines a cochain, although generally not a cocycle. The ordinary cochain identity nevertheless gives

$$\delta b = U \smile \widetilde{\delta h^*T}.$$

If its front U -face is zero, the evaluation is zero. Otherwise the entire parent lies in O , and every face entering the remaining coboundary is an actual pullback of T . They cancel because $\delta T = 0$. This proves the assertion without discarding any extension-by-zero terms separately. Each block or skip face is contained in its parent image and has the same property.

In a contributing term the U -face first puts the whole rectangle in the closed intermediate tube, where h is defined and uniformly continuous on the compact parameter set. Its nonzero T -face cannot have its entire image in $\{\ell < L\}$, and hence contains a point with $\ell(h) \geq L$. Refining the factors so that $\ell \circ h$ oscillates by less than ε on each such rectangle proves the estimate. The tube conditions follow in the same way from the gap between support and intermediate radii. Include all homotopy intervals among the factors before refining. The finitely many conditions then hold simultaneously. All later subdivision prisms and block or skip faces stay in the original rectangles, so both the domain and length conditions persist. $\square$

We will always use such small chains. Boundary comparisons have the following concrete meaning. Subdivide a finite homology adequately, then apply the cap boundary formula. Its subdivided boundary is compared with any earlier adequate representative by the subdivision prism inside the original simplex images. Factor subdivisions are compared by crossing these prisms with the other factors. For a homotopy of evaluation or output maps, cross with its interval *before* capping and refine all factors, so the endpoint chains are the original shuffled constructions. This proves compatibility with refinements and finite fillings at the same whole-parent length bounds. A single subdivision depth on the entire loop space is neither chosen nor needed. The shuffle step counts below are applied to these factorwise refined product chains. Comparison prisms need not have that form; they are handled by the supported boundary formula instead.

5.3 The cut and the join

Fix $\varepsilon = 10^{-3}$. Choose a small cut radius η_c so that a short round arc at distance less than $4\eta_c$ has g -length less than ε . The diagonal Thom class has a normalized representative

$$U_c \in C^n(M^2, F_{\eta_c} \cup \Delta M). \quad (5.6)$$

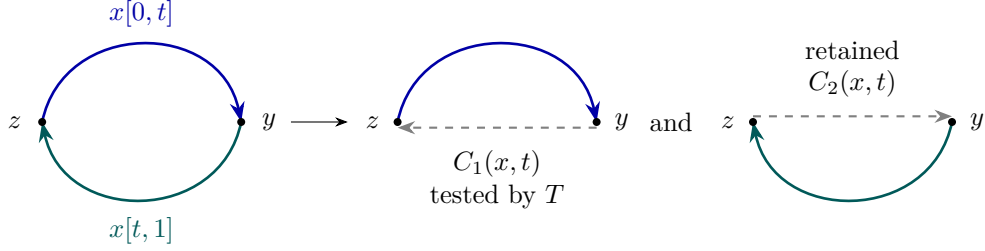


Figure 1: The native cut at $y = x(t)$, with $z = x(0)$. Short round arcs (dashed) close the prefix and the remaining path. The cut tests the first loop and retains the second. The drawing separates the nearby endpoints to show the two arcs.

To see this, its restriction to ΔM is the mod-two Euler class of TM , which vanishes because $\chi(S^n)$ is even. The exact sequence of the triple $F_{\eta_c} \subset F_{\eta_c} \cup \Delta M \subset M^2$ therefore lifts the Thom class to this relative group. A relative cocycle representative, followed by normalization, gives (5.6).

For paths γ, ζ with matching endpoints, let $\gamma \# \zeta$ denote concatenation with half the time assigned to each. Brackets $[z, z']$ denote a short round arc. On the cut tube define

$$\begin{aligned} C_1(x, t) &= x[0, t] \# [x(t), x(0)], \\ C_2(x, t) &= [x(0), x(t)] \# x[t, 1], \end{aligned} \quad (5.7)$$

where restrictions of a path are rescaled to the unit interval. Figure 1 shows the tested prefix and retained complement, with the short arcs which close them. These maps are jointly continuous in H^1 , including $t = 0, 1$. Indeed the derivative energy of a restriction to $[s, t]$ rescaled to unit time is $(t - s) \int_s^t |x'|^2$, which tends to zero at a collapsing interval, uniformly along an H^1 -convergent family.

For a normalized p -cocycle T vanishing on $\mathcal{A}_L$, put $q = n + p - 1$ and define on adequate chains

$$D_T x = (C_2)_{\#}((x \times I) \frown U_c(x(0), x(t)) \frown T(C_1(x, t))). \quad (5.8)$$

Pullback notation in this formula is abbreviated; the second cap is defined on the supported cut tube. Absolute extensions of the loop coproduct and their Euler-characteristic boundary correction were studied by Cieliebak, Hingston and Oancea [9, Section 8.1]. Their odd-sphere calculation also exhibits constant-loop output terms [9, Proposition 8.3 and Remark 8.4]. The filtered comparison for the cut defined here is proved below.

For joining, choose an arbitrarily small radius η_j and a normalized diagonal Thom cocycle U_j relative to F_{η_j} . For close z, z' , let $J_{z, z'}$ be the shortest plane rotation taking z to z' , and the identity at equality. The rotations $J_{z, z'}^u$, $-1 \leq u \leq 1$, depend smoothly on the two points and converge smoothly to the identity as the points approach. Their g -length distortion is at most $1 + \kappa$, with $\kappa \rightarrow 0$ as $\eta_j \rightarrow 0$. Define

$$\mu(x, y) = ((J_{x(0), y(0)} x) \# y)_{\#}((x \times y) \frown U_j(x(0), y(0))). \quad (5.9)$$

Lemma 5.3 (Boundary and length properties). *The cut has degree $-q$, takes absolute cycles to absolute cycles, and preserves global null-homology. For a cycle in $\mathcal{A}_a$ it is supported in*

$$\mathcal{A}_{a-L+1}. \quad (5.10)$$

The same cut estimate holds on finite fillings. The join has degree $-n$ and satisfies the product boundary formula. On any prescribed finite collection of chains and fillings its radius can be

chosen so that its output length is less than the sum of the two input length bounds plus 1. All these statements are compatible with the refinement comparisons above.

If c is a fundamental cycle of constant loops, then

$$\mu(c, x) \simeq x \quad \text{in } \mathcal{A}_a \quad (5.11)$$

for every cycle x in $\mathcal{A}_a$.

Proof. The cut degree is $1 - n - p = -q$. The cap boundary formula applies by Lemma 5.2; its time-end terms vanish because $(x(0), x(t))$ is exactly diagonal at $t = 0, 1$. Applying the same construction to a finite filling proves null-homology preservation. On a surviving parent the tested first loop has length at least $L - \varepsilon$, while

$$\ell(C_1(x, t)) + \ell(C_2(x, t)) \leq \ell(x) + 2\varepsilon.$$

This gives (5.10). If its upper level is nonpositive, no term survives. The join boundary formula uses the closed cocycle U_j . Its output length is at most $(1 + \kappa)\ell(x) + \ell(y)$, and η_j may be chosen after all finite length bounds have been specified.

For the unit, consider a shuffle of $c \times x$. A constant-factor step remaining after the n -cap makes the output degenerate: the rotated constant is exactly $x(0)$, so the output is independent of that factor. The remaining terms have all n constant-factor steps first. Their coefficient is the Thom evaluation on $[M] \times \{x(0)\}$, at the initial parameter vertex of the x -simplex, and equals one. Thus the output is a constant half followed by x . Removing the waiting half by reparametrization gives (5.11) without changing length. $\square$

5.4 Interchanging which factor is cut

Theorem 5.4 (Filtered interchange). *Let T be a normalized cocycle of degree $p \geq 1$ vanishing on $\mathcal{A}_L$, where $L > 0$, and put $q = n + p - 1$. Let x and y be arbitrary absolute cycles of respective degrees j, N with*

$$j > n + p, \quad N > n + p,$$

supported in $\mathcal{A}_a$ and $\mathcal{A}_b$. With sufficiently small join radius and adequate refinements,

$$\mu(D_T x, y) \simeq \mu(D_T y, x) \quad \text{in } \mathcal{A}_{a+b-L+c_{cj}}, \quad c_{cj} = 4. \quad (5.12)$$

Both sides have degree $j + N - q - n$. The constant c_{cj} is independent of both degrees, the cocycle, and all length bounds. The join radius may be chosen for any additional prescribed finite collection of fillings without changing this constant.

Proof. Write $J = J_{x(0), y(0)}$ and $Z = (Jx) \# y$. On $x \times I_s \times y$, first cap with $\sigma = U_j(x(0), y(0))$, then with $U_c(Z(0), Z(s))$, then with $T(C_1(Z, s))$, and output $C_2(Z, s)$. Call this cycle $P(x, y)$. Its time endpoints vanish by (5.6). We prove

$$\mu(D_T x, y) \simeq P(x, y) \simeq P(y, x) \simeq \mu(D_T y, x). \quad (5.13)$$

All homologies will have the bound in (5.12).

The second half vanishes. Split I_s at $1/2$. At the seam $Z(1/2) = Z(0)$, so each piece is separately a cycle after capping. For $s \geq 1/2$, both the cut evaluation and the second-cut output depend only on (y, s) . A shuffle contains j steps from x . Its join face uses at most n of them, and its T -face at most p . Since $j > n + p$, another such step belongs either to the cut Thom face or to the retained output face. In the first case the normalized Thom evaluation is zero; in the second the output simplex is degenerate. In fact projection drops the unused x -step, and the output factors through the corresponding standard degeneracy map. Deleting a repeated projected vertex uses a face of the original simplex, so this factorization stays within the cut domain. The second-half cycle therefore bounds in its image sublevel.

The first half gives the native cut. Put $t = 2s$ on the first half. The first cut is exactly $JC_1(x, t)$. The second cut consists of an arc, the rotated x -tail and y , with respective durations

$$\frac{1}{2}, \quad \frac{1-t}{2(2-t)}, \quad \frac{1}{2(2-t)}.$$

Interpolate them linearly to $1/4, 1/4, 1/2$. The result is $(JC_2(x, t))\#y$. Length is unchanged, and continuity at $t = 1$ follows because the shrinking tail contributes at most a fixed multiple of $\int_t^1 |x'|^2$ to the energy.

Keep this output fixed and move the cut evaluation and tested first loop from those of Jx to those of $J^u x$, with u decreasing from 1 to 0. Cross with this parameter before capping. Round rotations preserve the cut distance, so all surviving parents support the fixed output throughout. The cap boundary formula gives a homology: it requires continuous supported maps, not an identity between the fixed output and an intermediate tested cut. Its time endpoints remain diagonal. Small rotation distortion keeps the first-cut plus output length within a fixed small error of $\ell(x) + \ell(y)$.

At $u = 0$ put

$$\gamma = U_c(x(0), x(t)) \smile T(C_1(x, t)),$$

using extension by zero as in Lemma 5.2. This normalized cochain has degree $n + p$ and is independent of y . Impose the native cut smallness even before the join cap. We may then commute σ and γ using

$$\delta(\sigma \smile_1 \gamma) = \sigma \smile \gamma + \gamma \smile \sigma + \sigma \smile_1 \delta\gamma. \quad (5.14)$$

For completeness, if the degrees of cochains u, v are r, s , their mod-two cup-one on a simplex of dimension $r + s - 1$ is

$$\sum_{i=0}^{r-1} u[0, \dots, i, i+s, \dots, r+s-1] v[i, \dots, i+s].$$

The coboundary formula follows by pairing deletions outside and inside the block; the shared-end deletions cancel in adjacent summands except for the two outer cups. This gives (5.14) since $\delta\sigma = 0$.

Lemma 5.2 makes $\delta\gamma$ zero on every block involved, including all extension-by-zero terms. Each nonzero summand of the primitive has a join face and a cut/test block. Its parent is therefore in both tubes and has high first-cut length. On either time endpoint the cut part of every γ -face vanishes. Capping with the primitive and pushing to the fixed output supplies the required homology between the two cap orders.

With γ first, a y -step in its front face gives a degenerate evaluation. The surviving shuffles start with $n + p$ steps from $x \times I$. Their suffixes are exactly the shuffles of the retained cut face with y . The native second cut starts at $x(0)$, so the remaining join cap and output are precisely those of $\mu(D_T x, y)$. Common factor refinements compare any independently chosen join subdivision. This proves the first comparison in (5.13).

Symmetry of the joined cut. Keep the cut interval whole during the following homotopies. Rotate the entire loop by J^{-u} , obtaining

$$(J^{1-u}x)\#(J^{-u}y), \quad 0 \leq u \leq 1.$$

Then translate the circle parameter of $x\#(J^{-1}y)$ through half a period, ending at $(J^{-1}y)\#x$. Translation is jointly continuous in H^1 and preserves length. At every stage the evaluations at cut times zero and one are diagonal. An intermediate half-time seam need not be diagonal; any seam terms cancel because we use the full interval.

The swapped join cochain represents the same Thom class relative to F_{η_j} , and hence differs from σ by the coboundary of a cochain relative to that far set. A nonzero face of this primitive forces its whole adequate parent into the join tube. Cap with the primitive followed by the cut and test cochains and use the boundary formula; the cut kills the time endpoints. Shuffle symmetry now identifies the result with $P(y, x)$. Repeating the first-half argument in the opposite order proves the last comparison in (5.13); the second half vanishes because $N > n + p$.

One length bound for all these homologies. Choose the join radius so that each partial rotation changes the length of every path with the prescribed finite bounds by less than ε . This is possible after including the bounds for any additional finite fillings. In the whole-loop homotopies a surviving test saves at least $L - \varepsilon$ from the common joined length, at a cost of two short arcs and small rotations. In the first-half homotopy with fixed output, compare the tested $J^u C_1$ with C_1 ; each differs in length by less than ε . Together with the two cut arcs and the output rotation this gives a cost less than 8ε . Timing changes preserve length. The cup-one and primitive comparisons satisfy the same bound by whole-parent support and oscillation. Thus each individual homology lies already below $a + b - L + 1$ after sufficiently small choices. Their sum, together with the image-preserving refinement and normalization homologies, satisfies the generous common bound $a + b - L + 4$. The number of chain terms does not add lengths: the support of a sum is contained in the union of their supports. Neither a larger degree nor a finer subdivision traverses an additional long loop. This proves the claimed uniformity. $\square$

5.5 Relative scalar comparison and the absolute unit

The remaining identification uses the established singular models of the two string-topology products. We state the scalar comparison separately because it does not determine constant-loop output components.

Lemma 5.5 (Comparison with the string-topology products). *The join μ induces $\bullet$ on ordinary homology. If T represents a relative class $[T] \in H^p(\Lambda, M)$ and b is a normalized relative cocycle of degree r , then for every absolute cycle z of degree $p + r + n - 1$,*

$$\langle b, D_T z \rangle = \langle [T] \otimes [b], [z] \rangle. \quad (5.15)$$

Here the brackets on the right also denote the image of z in relative homology.

Proof. Hingston–Wahl give singular chain models on H^1 loops by capping with the evaluation pullback of a diagonal Thom cocycle, then adding short arcs and concatenating or cutting [20, Theorem A, Definitions 1.2 and 1.4]. The dual relative operation is the Goresky–Hingston product [20, Theorem 2.13]; all sign differences disappear over $\mathbb{F}_2$. We use these models with the auxiliary round metric and compare the actual maps and cochains. Choose their support radius to equal ours and their map domain to contain our larger tube; both choices can be made within their injectivity-radius requirement.

For joining, partially rotate the first input by J^u and apply their short-arc retraction, whose values have equal starts throughout. Writing $z = x(0)$ and $z' = y(0)$, the residual mismatch satisfies $d_{\text{rd}}(J^u z, z') = (1 - u)d_{\text{rd}}(z, z')$, so all inputs remain in its domain. At $u = 0$ this is their retraction; at $u = 1$ the inputs already have equal starts, so their retraction is the identity and gives our rotation retraction. This is a homotopy inside equal-start pairs. Interpolate their energy-optimal concatenation times to halves. For path pieces of square-root energies P, Q , the optimal time for the first piece is $P/(P + Q)$; during this interpolation its

energy contribution is bounded by

$$\max\{2P^2, P(P+Q)\}.$$

This tends to zero if that piece collapses. The other piece is handled in the same way, and if both collapse both contributions vanish. Thus the timing homotopy is continuous also at constant pieces.

For cutting, their short-arc retraction followed by cutting gives exactly the two paths in (5.7), with optimal instead of half timings. The same interpolation compares them. At either time endpoint one of the two paths remains constant. Pairing with T and b evaluates on $z \times I$ the cochain

$$U_c(x(0), x(t)) \smile \chi, \quad \chi = T(C_1(x, t)) \smile b(C_2(x, t)).$$

Both tests are relative to constants. Hence χ vanishes at $t = 0$ through its first factor and at $t = 1$ through its second factor.

Replace U_c by the reduced ordinary Thom representative used in that model. They differ by a coboundary relative to the far set. Although its primitive need not vanish on the diagonal, all of its time-end contributions are multiplied by χ and vanish. If a cap convention reverses the cup order, the cup-one formula supplies the comparison. On a small simplex in the tube, χ is a closed pullback; on a simplex in the far set, every face of the accompanying Thom cochain or relative primitive vanishes. Thus extension-by-zero errors contribute nothing, exactly as in Lemma 5.2. The same argument permits replacement of either relative test by a cohomologous representative. Join Thom representatives are compared in the same way, without interval-end terms.

Finally the Alexander–Whitney map in the coproduct evaluates $T \otimes b$ as the displayed cup of the two pullbacks. Their small-chain replacement and our refined products represent the same relative chain class: both are homologous to the input, and small-chain inclusion is a quasi-isomorphism, also relative to the time ends, by subdivision. This proves (5.15) and the asserted join identification. The argument compares two relative tests; it makes no assertion about a canonical absolute extension of the coproduct. $\square$

Proposition 5.6 (The cut of the power is the absolute unit). *For the choices S, T, l, L in (5.4),*

$$[D_T S] = [M] \quad \text{in } H_n(\Lambda). \quad (5.16)$$

Moreover $D_T S$ is supported in $\mathcal{A}_{l-L+1}$, and there is a constant B independent of m such that it is homologous to a fundamental constant-loop cycle in $\mathcal{A}_B$.

Proof. Represent $\Theta^{\bullet(m+1)}$ by a finite cycle z . Its cut has degree $n + 2(m+1)d - q = 3n - 2$. Lemma 5.5, commutativity of the relative product over $\mathbb{F}_2$, and (5.3) give $\langle \theta, D_T z \rangle = 1$. The rank-one calculation therefore yields $[D_T z] = \Theta$.

Both $|z|$ and $|S| = n + q$ exceed $n + p = q + 1$. Apply Theorem 5.4 to these cycles, now forgetting the finite output length bound. Using the join identification,

$$0 \neq \Theta^{\bullet(m+1)} = [\mu(D_T z, S)] = [\mu(D_T S, z)].$$

If $D_T S$ were null-homologous, its join with z would bound, by Lemma 5.3. Thus $D_T S \neq 0$ in $H_n(\Lambda) = \mathbb{F}_2[M]$, proving (5.16). This determines the absolute constant component by interchange and nonvanishing, rather than by the relative scalar test alone.

The cut bound gives support in $\mathcal{A}_{l-L+1}$. Its difference from a constant fundamental cycle is globally null in degree n ; Proposition 4.6 fills that difference in $\mathcal{A}_{l-L+1+D_n}$. By (5.5), this is one fixed level B . $\square$

The canonical absolute coproduct of [20, Theorem B] removes evaluation components; the cut used here retains the component determined in Proposition 5.6. The comparison proved above is specific to this cut and its degree range. It is not an invocation of a general Frobenius identity; the canonical extension fails such an identity [20, Remark 4.14].

We have obtained three compatible pieces of information: a cut which lowers degree and saves length, an interchange homology with fixed length cost, and a bounded-length correction of $D_T S$ to the unit. The next section combines them with fixed-degree filling to remove the dependence of the filling increment on the homological degree.

6 One filling increment in every degree

We combine fixed-degree shortening with the filtered cut-join identity. The cut removes a degree-dependent amount of length, while its join with a sphere power restores that length up to one fixed difference. The result is a filling increment independent of both degree and level.

Theorem 6.1 (Uniform ordinary death). *For every smooth Riemannian metric on S^n , $n \geq 3$, there is a constant $D > 0$ such that, for every integer $j \geq 0$ and every real $a > 0$,*

$$\ker(H_j(X_a) \longrightarrow H_j(\Lambda)) \subseteq \ker(H_j(X_a) \longrightarrow H_j(X_{a+D})), \quad (6.1)$$

$$\ker(H_j(X_a, M) \longrightarrow H_j(\Lambda, M)) \subseteq \ker(H_j(X_a, M) \longrightarrow H_j(X_{a+D}, M)). \quad (6.2)$$

The same constant works at regular and nonregular levels. No assumption on the number of prime geodesic images is required.

Proof. We first prove the absolute assertion for the length filtration $\mathcal{A}_a$, then pass to square-root energy and to the relative pair.

Choosing the degree reduction. Let $d = n - 1$ and let α, C_H be the constants of Proposition 5.1. For sufficiently large j put

$$m = \left\lfloor \frac{j-2}{2d} \right\rfloor, \quad q = 2md, \quad p = q - d, \quad N = n + q. \quad (6.3)$$

Then

$$2 \leq r := j - q \leq 2d + 1 = 2n - 1, \quad j > q + 1 = n + p, \quad N > n + p. \quad (6.4)$$

Choose an integer $J > 2n$ so large that, whenever $j \geq J$, these choices have $m \geq 1$, $p > n + 1$, and $L > 0$ in (5.4). Take S, T, l, L as in that equation. Write

$$C_{\text{gap}} = l - L = \frac{2n - 1 + 2C_H}{\alpha} + 4, \quad D_* = \max_{0 \leq s < J} D_s, \quad (6.5)$$

where D_s is furnished by Proposition 4.6. All the remainder degrees in (6.4), and the degree n , occur in this finite maximum.

Filling a high-degree null cycle. Let x be a finite globally null j -cycle in $\mathcal{A}_a$, with $j \geq J$. Its cut $D_T x$ is a globally null r -cycle in $\mathcal{A}_{a-L+1}$ by Lemma 5.3. If $a - L + 1 > 0$, Proposition 4.6 gives a chain W such that

$$\partial W = D_T x, \quad \text{supp } W \subset \mathcal{A}_{a-L+1+D_r}.$$

If $a - L + 1 \leq 0$, the cut has no surviving term and we take $W = 0$.

Fix a fundamental constant-loop cycle c . By Proposition 5.6, the cycle $D_TS - c$ is globally null in degree n and is supported in $\mathcal{A}_{C_{\text{gap}}+1}$. Fixed-degree filling gives a chain V with

$$\partial V = D_TS - c, \quad \text{supp } V \subset \mathcal{A}_{C_{\text{gap}}+1+D_n}. \quad (6.6)$$

This length bound is independent of j , although the chain V itself may depend on the chosen power.

Choose the join radius after these finite length bounds have been specified. Lemma 5.3 and Theorem 5.4 permit one choice for the interchange homology and both joined fillings. Refinement comparisons are included in the same bounds. The relevant chains have the following supports:

$$\mu(W, S) \text{ lies in } \mathcal{A}_{a+C_{\text{gap}}+D_*+2}, \quad (6.7)$$

$$\mu(V, x) \text{ lies in } \mathcal{A}_{a+C_{\text{gap}}+D_*+2}. \quad (6.8)$$

The first assertion is omitted when $W = 0$; otherwise it follows by adding the input bounds $a - L + 1 + D_r$ and l and the join allowance 1. The second follows from (6.6) in the same way. Both chains have degree $j + 1$.

The degree inequalities (6.4) are precisely the two hypotheses needed for the interchange homology. It gives

$$\mu(D_Tx, S) \simeq \mu(D_TS, x) \quad \text{in } \mathcal{A}_{a+C_{\text{gap}}+4}.$$

The chain $\mu(W, S)$ has boundary $\mu(D_Tx, S)$, while $\mu(V, x)$ has boundary $\mu(D_TS, x) - \mu(c, x)$. Finally (5.11) joins $\mu(c, x)$ to x in $\mathcal{A}_a$. Adding these chains and the interchange homology therefore fills x .

Thus one permissible length increment is

$$D = C_{\text{gap}} + D_* + 6. \quad (6.9)$$

It exceeds every bound just used. For $j < J$, the conclusion follows directly from Proposition 4.6, since $D_j \leq D_* < D$. We have proved that every globally null finite cycle in $\mathcal{A}_a$, in any degree, fills in $\mathcal{A}_{a+D}$. The constants in (6.9) depend only on the metric and n , and every construction used above is independent of a finite-prime hypothesis.

Returning to square-root energy. A cycle x in X_a is in $\mathcal{A}_a$ because $\ell \leq F$. Fill it by the length assertion and apply Ψ_1 to that filling. Lemma 4.1 gives $F(\Psi_1\gamma) = \ell(\gamma)$, so the new filling lies in X_{a+D} and has boundary Ψ_1x . The normalization prism from x to Ψ_1x lies entirely in X_a by (4.1). Adding this prism proves (6.1) with the same D . Strict sublevel inequalities are preserved throughout; no approximation by regular levels is involved.

Passing to constants-relative homology. Let $i : M \rightarrow \Lambda$ include constants and let $\text{ev}_0 : \Lambda \rightarrow M$ be evaluation at zero. Since $\text{ev}_0 \circ i = \text{id}_M$, the chain map

$$P = \text{id} - i_{\#}(\text{ev}_0)_{\#}$$

annihilates $C_*(M)$ and preserves $C_*(X_a)$ for every $a > 0$. If z represents a relative cycle in (X_a, M) , then $\partial z \in C_*(M)$, so Pz is an absolute cycle representing the same relative class. If z is globally null relative to M , there are finite chains U in Λ and c' in M with $\partial U = z + c'$. Applying P gives $\partial(PU) = Pz$. Hence Pz is globally null absolutely. By (6.1) it fills in X_{a+D} , and passage to the relative chain quotient proves (6.2). This argument includes degree zero and uses exactly the same increment. $\square$

The restriction $a > 0$ ensures that the constant loops belong to the relative pair. For nonpositive a the open absolute sublevel is empty. Theorem 6.1 supplies the degree-uniform control of ordinary transient classes needed for the reflection-equivariant argument below.

7 Global equivariant classes

We now compute the global reflection cohomology and construct the full-group classes needed in the doubling argument. The computation uses a round metric only to filter the loop space; the resulting classes belong to the same global groups for every metric. The distinction between odd and even round stages will be essential. It is independent of the parity of the sphere dimension.

7.1 Natural Gysin sequences

Let $w \in H^1(BG; \mathbb{F}_2)$ and $U \in H^2(BG; \mathbb{F}_2)$ be the Stiefel–Whitney classes of the standard two-dimensional representation of $G = O(2)$. We use the same letter w for its restrictions. Let $v \in H^1(BK; \mathbb{F}_2)$ be the character which is nontrivial on τ and trivial on R . Then

$$H^*(BG; \mathbb{F}_2) = \mathbb{F}_2[w, U], \quad H^*(BK; \mathbb{F}_2) = \mathbb{F}_2[v, w], \quad \text{res}_K U = v(v + w). \quad (7.1)$$

Indeed, the standard plane restricts to the two characters v and $v + w$. Its projectivization over BG is BK , so the projective-bundle formula makes restriction injective [17, Section 3.1, pp. 78–79]. Conjugation in G interchanges the two coordinate lines of K , whereas inner conjugation acts trivially on $H^*(BG)$. The image is consequently contained in the symmetric polynomials in $v, v + w$. It contains their sum w and product U , which generate that symmetric polynomial ring. This proves (7.1).

For an H -pair (Y, A) , write $H_H^*(Y, A)$ for its Borel cohomology. These groups are zero in negative degrees.

Proposition 7.1 (Natural Gysin sequences). *For every G -pair (Y, A) , there are natural exact sequences*

$$\cdots \longrightarrow H_G^{j-2}(Y, A) \xrightarrow{U} H_G^j(Y, A) \xrightarrow{\text{res}_R} H_R^j(Y, A) \xrightarrow{\int} H_G^{j-1}(Y, A) \xrightarrow{U} H_G^{j+1}(Y, A) \longrightarrow \cdots, \quad (7.2)$$

$$0 \longrightarrow H_G^j(Y, A) \xrightarrow{\text{res}_K} H_K^j(Y, A) \xrightarrow{\int} H_G^{j-1}(Y, A) \longrightarrow 0. \quad (7.3)$$

For an R -pair and a K -pair, respectively, the sign-line sequences are

$$\cdots \longrightarrow H_R^{j-1}(Y, A) \xrightarrow{w} H_R^j(Y, A) \longrightarrow H^j(Y, A) \longrightarrow H_R^j(Y, A) \xrightarrow{w} H_R^{j+1}(Y, A) \longrightarrow \cdots, \quad (7.4)$$

$$\cdots \longrightarrow H_K^{j-1}(Y, A) \xrightarrow{v} H_K^j(Y, A) \longrightarrow H^j(Y, A) \longrightarrow H_K^j(Y, A) \xrightarrow{v} H_K^{j+1}(Y, A) \longrightarrow \cdots. \quad (7.5)$$

The maps commute with maps of pairs and subgroup restriction wherever the latter applies, and satisfy the projection formula for absolute base classes.

Proof. For a real rank- r representation of H , its associated bundle over Y_H restricts to a bundle over A_H . Suppose that its unit sphere is H/H' . The sphere-bundle pair is then the Borel pair for H' . The relative Thom sequence gives

$$\cdots \longrightarrow H_H^{j-r}(Y, A) \xrightarrow{w_r} H_H^j(Y, A) \longrightarrow H_{H'}^j(Y, A) \xrightarrow{\int} H_H^{j-r+1}(Y, A) \longrightarrow \cdots. \quad (7.6)$$

Here is a cochain construction that also justifies (7.6) when the inclusion $A_H \subset Y_H$ is not a cofibration. For a vector bundle over $Z \supset A_0$, let D, S denote its disk and sphere bundles, and let D_{A_0}, S_{A_0} be their restrictions. In the singular chain complex of D , put

$$\mathcal{L} = C_*(S) + C_*(D_{A_0}).$$

The two summands intersect in $C_*(S_{A_0})$, since singular chains have the singular simplices as a basis. Restriction of cochains gives degreewise exact sequences

$$\begin{aligned} 0 \longrightarrow \text{Ann } \mathcal{L} &\longrightarrow C^*(D, S) \longrightarrow C^*(D_{A_0}, S_{A_0}) \longrightarrow 0, \\ 0 \longrightarrow \text{Ann } \mathcal{L} &\longrightarrow C^*(D, D_{A_0}) \longrightarrow C^*(S, S_{A_0}) \longrightarrow 0. \end{aligned}$$

Surjectivity follows by extending linear functionals on chain subspaces. Cup with a Thom cocycle sends a relative base cochain into $\text{Ann } \mathcal{L}$: on S the Thom factor vanishes, and on D_{A_0} the base factor vanishes. Comparison of the first sequence with the base-pair sequence, using the absolute Thom isomorphisms, proves

$$H^{j-r}(Z, A_0) \cong H^j(\text{Ann } \mathcal{L}).$$

The second sequence and the disk contraction give (7.6). The map back to the base is cup with the zero-section pullback of the Thom class, namely w_r . Pullback of Thom classes gives naturality and the projection formula. The absolute Thom isomorphism used here is the ordinary mod-two one [17, Section 3.2, pp. 88–89]; for a general base it follows by pulling back to its singular CW replacement and using the disk and sphere fibrations. This argument uses the sum of two chain subspaces and does not require an excision assertion for their union.

The standard G -plane has sphere stabilizer R and Euler class U , proving (7.2). Let ρ_2 be the representation that doubles rotation angles and sends the chosen reflection to $\text{diag}(1, -1)$. Its sphere stabilizer is K . On K it has a fixed line, so its Euler class restricts to zero. The injectivity in (7.1) makes that Euler class zero already on BG , giving (7.3). Finally, the sign representations with kernels $1 \subset R$ and $R \subset K$ have Euler classes w and v , respectively. $\square$

7.2 The classes to be constructed

Evaluation at zero is R -invariant. If $\sigma \in H^n(S^n; \mathbb{F}_2)$ is the nonzero class, set

$$e = \text{ev}_0^* \sigma \in H_R^n(\Lambda; \mathbb{F}_2).$$

In particular $e^2 = 0$. Recall that $d = n - 1$, and put

$$a_m = (2m - 1)d \quad (m \geq 1).$$

Theorem 7.2 (Global generators and full-group lifts). *For $n \geq 3$, there are classes*

$$X_m \in B_\infty^{a_m} = H_R^{a_m}(\Lambda, M), \quad m \geq 1,$$

with the following properties.

(i) *The classes*

$$\xi_{m,k} = w^k X_m, \quad \eta_{m,k} = e w^k X_m, \quad m \geq 1, \quad 0 \leq k \leq d,$$

form a graded $\mathbb{F}_2$ -basis of B_∞ . Their degrees are

$$|\xi_{m,k}| = a_m + k, \quad |\eta_{m,k}| = a_m + n + k.$$

We call the $\xi_{m,k}$ the bottoms and the $\eta_{m,k}$ the tops at stage m . Multiplication by w^n annihilates B_∞ .

(ii) *Every X_m is the restriction of a class $\zeta_m \in H_G^{a_m}(\Lambda, M)$. If m is odd, then $U^n \zeta_m = 0$. If m is even, there is also a class $Y_m \in H_G^{a_m+n}(\Lambda, M)$ with $\text{res}_R Y_m = e X_m$.*

- (iii) For the round metric, let Z_p be an open regular sublevel containing exactly the first p positive critical manifolds, and choose Z_0 below the first one. For $H = R, G$, every crossing sequence for (Z_p, Z_{m_0-1}) , $p \geq m_0$, is short exact. A chosen class of the m -th crossing has a unique extension to $H_H^j(\Lambda, Z_{m-1})$ whenever $j < a_{m+1}$, for $H = R, G$. The lifts in (ii) and the classes X_m can be obtained from these relative extensions.

The top classes in (ii) are chosen lifts; they need not be canonical. Only their reflection restrictions will be used. Also, the annihilation statement in (i) is relative to constants. The absolute constant-loop Borel summand retains its polynomial w -tower.

7.3 Round crossings

Give S^n its unit round metric and write E_0 for its energy. The positive critical values are $c_m = (2\pi m)^2$. The critical manifold at c_m consists of the loops

$$\gamma_{x,u}(t) = x \cos(2\pi mt) + u \sin(2\pi mt), \quad |x| = |u| = 1, \quad x \perp u. \quad (7.7)$$

It is $C_m = STS^n$, the manifold of orthonormal two-frames in $\mathbb{R}^{n+1}$.

Lemma 7.3 (Round attachment). *The negative bundle of C_m has rank a_m , and the kernel of the round Hessian is exactly TC_m . For every closed subgroup $H \subset G$, the crossing has a Thom isomorphism*

$$H_H^j(Z_m, Z_{m-1}) \cong H_H^{j-a_m}(C_m). \quad (7.8)$$

These isomorphisms commute with subgroup restriction and cup by absolute classes. The small sublevel Z_0 retracts G -equivariantly to M .

Proof. Along (7.7), put

$$e_t = -x \sin(2\pi mt) + u \cos(2\pi mt).$$

A tangent field is $a(t)e_t + b(t)$, where $b(t) \perp \text{span}\{x, u\}$. Half the Hessian is

$$\int_0^1 (|a'|^2 + |b'|^2 - (2\pi m)^2 |b|^2) dt. \quad (7.9)$$

In each of the d normal directions, the negative Fourier frequencies are $|k| < m$. Their real dimension is $2m - 1$. The null modes are the two normal modes at frequency m , together with the constant tangential mode. They have dimension $2d + 1$ and are precisely the variations of the frame (x, u) . The nonzero Fourier eigenvalues, divided by the corresponding squared H^1 -norms, are bounded away from zero. Thus the Hessian is invertible on the orthogonal complement of its kernel. At constants, the Hessian is $2 \int |V'|^2$, positive and invertible off its constant kernel.

We describe why the resulting attachment respects the full group. Use the strong metric induced by the Euclidean H^1 -norm. Each parameter symmetry is a smooth isometry, and the action is continuous. Pointwise exponential gives a chart on the Hilbert normal bundle of the compact manifold C_m , at a uniform small radius. In these coordinates Taylor's formula writes the energy difference as $\langle A(z, y)y, y \rangle$, with $A(z, 0)$ an invertible symmetric operator on the normal fiber. Normalize first by $|A(z, 0)|^{1/2}$, and let $J = \text{sign } A(z, 0)$. For the resulting operator B near J , the norm-convergent power series $C = (JB)^{1/2}$ satisfies $C^*JC = B$. The change $y \mapsto C(z, y)y$ is a diffeomorphism after shrinking the tube. All these constructions commute with the parameter isometries. They give the exact normal form

$$E_0 = c_m - |u_-|^2 + |u_+|^2 \quad (7.10)$$

on the negative and positive bundles.

For completeness, the sublevel attachment uses the original Hilbert space, rather than a polygon model with only finite rotational symmetry. Patch a complete normalized descending gradient field to the split field

$$\dot{z} = 0, \quad \dot{u}_- = u_-, \quad \dot{u}_+ = -u_+$$

inside the normal tube, using an invariant squared-radius cutoff. Both fields strictly decrease E_0 away from C_m within the isolating energy band. The coordinate derivatives are uniformly bounded in a small tube over the compact zero section, so the patched field remains bounded. The ambient strong metric is complete; hence this field is complete. Palais–Smale and regular-band deformation (Lemmas 3.1 and 3.2) give a uniform decrease outside any smaller tube in an isolating energy band. The flow-lens construction of Lemma 3.6 therefore computes the crossing in that smaller tube. Use its notation X, Y, N, Q , with lower energy cut $t_- < c_m$. In split coordinates, contraction of u_+ decreases both endpoint energies of a trapped flow segment. It defines maps $N \rightarrow X$ taking Q into Y , even when a contracted point leaves N through the lower endpoint. Via $N/Q \cong X/Y$, these maps induce a contraction to the negative disk with its boundary collapsed. For flow time T , the surviving disk before applying the flow has radius $e^{-T}\sqrt{c_m - t_-}$; its boundary flows to the lower cut. The flow multiplies u_- by e^T , identifying this disk and sphere with the usual negative-bundle disk and sphere. The disk is closed in the lens and meets Q exactly in its sphere. These statements remain true after Borel construction. The disk/sphere quotient consequently embeds in the lens quotient: the saturation of a closed subset of the disk is that subset, possibly together with the closed collapse set. The contraction fixes its image. The disk/sphere inclusion is a good pair. This proves (7.8) by the mod-two Thom isomorphism.

All pair maps just used are G -equivariant. The negative bundle carries its actual derivative action; no triviality of that action is assumed. Restriction to a subgroup and cup by an absolute class therefore commute with the attachment isomorphism. Wider regular cuts give the same assertion by regular-band deformation. Finally, sufficiently small energy places a loop in the positive normal chart of the constants, where contraction of the normal coordinate proves the last assertion. $\square$

7.4 Cohomology of the critical manifolds

Reflection sends (x, u) to $(x, -u)$. It acts freely on C_m , and

$$(C_m)_R \simeq P(TS^n).$$

The sign line is the tautological line in this projective bundle. Since $TS^n \oplus \mathbb{R}$ is trivial, its positive Stiefel–Whitney classes vanish. The projective-bundle formula gives

$$H_R^*(C_m) = \mathbb{F}_2[w, e]/(w^n, e^2). \quad (7.11)$$

Here e is pulled back from the first frame vector x . This computation uses mod-two coefficients also when n is even. Write x_m for the bottom Thom class of the crossing, with the group understood. Its subgroup restrictions are the corresponding bottom classes. Thus the reflection crossing has basis

$$w^k x_m, \quad ew^k x_m, \quad 0 \leq k \leq d, \quad (7.12)$$

with degrees $a_m + k$ and $a_m + n + k$.

For $j \geq 1$, let $\rho_j : G \rightarrow O(2)$ multiply rotation angles by j and preserve the chosen reflection. The pullbacks of the two plane classes are

$$w_1(\rho_j) = w, \quad w_2(\rho_j) = (j \bmod 2)U. \quad (7.13)$$

To check the second identity, restrict to $BSO(2)$, where a weight- j complex line multiplies the first Chern class by j . The only possible additional degree-two term is a multiple of w^2 ; restriction to BR , where the plane has a trivial line, excludes that term.

Lemma 7.4 (Odd and even critical modules). *Define homogeneous polynomials $A_i \in \mathbb{F}_2[w, U]$ by*

$$A_0 = 1, \quad A_1 = w, \quad A_i = wA_{i-1} + UA_{i-2} \quad (i \geq 2). \quad (7.14)$$

If m is odd, the G -crossing is generated over $\mathbb{F}_2[w, U]$ by x_m , and

$$A_{d+1}x_m = A_{d+2}x_m = 0. \quad (7.15)$$

If m is even, multiplication by U is injective on the G -crossing, and that crossing is generated over $\mathbb{F}_2[w, U]$ by x_m and a chosen class y_m of degree $a_m + n$ whose reflection restriction is ex_m .

Proof. The G -action on frames is the right action through ρ_m , up to the convention for inverse precomposition. For odd m , the map from the critical Borel space to $\text{Gr}_2(\mathbb{R}^{n+1})$ has fiber the classifying space of the cyclic group of order m . Local choices of frames give local trivializations. That fiber has no positive-degree mod-two cohomology, by transfer, so the map induces a cohomology isomorphism. The tautological plane pulls back to the plane of ρ_m ; its classes are w, U by (7.13).

These two classes generate the Grassmann cohomology. Indeed, projectivize the tautological plane. On the resulting ordered-line flag space, let s, t be the first Stiefel–Whitney classes of its two lines. Over the Grassmannian, the projective-bundle formula gives a free module with basis $1, s$. Over $\mathbb{RP}^n$, the same flag space is the projectivization of the first line’s perpendicular bundle, so its ring is generated by s, t . Since $s + t = w$, $st = U$, and $s^2 = ws + U$, every polynomial in s, t reduces to $P(w, U) + sQ(w, U)$. Uniqueness of the free-module expansion implies that every class pulled back from the Grassmannian is a polynomial in w, U . The complementary bundle has rank d , with total Stiefel–Whitney class $(1 + w + U)^{-1}$. Its components are the polynomials A_i , proving (7.15) and the odd-stage assertion.

If m is even, τ acts trivially on the critical manifold. Consequently

$$H_K^*(C_m) = H_R^*(C_m)[v].$$

The same statement holds for the crossing modules after the Thom shift, with the actual equivariant negative bundle retained. The polynomial $U = v(v + w)$ is monic in v , hence acts injectively even though the coefficient ring in (7.11) has zero divisors. The injection in (7.3) makes U injective on the G -crossing as well. Sequence (7.2) therefore identifies the quotient of the G -crossing by U with the R -crossing. Choose a lift y_m of ex_m ; the bottom x_m already lifts. Their w -multiples generate modulo U , by (7.12). Subtracting such a combination and dividing the remainder by U reduces degree by two, so induction from the bottom degree proves generation. $\square$

7.5 Collapse on all round intervals

Lemma 7.5 (Interval collapse and supported extensions). *Fix $m_0 \geq 1$. For $H = R, G$ and $p \geq m_0$, the crossing sequence is short exact:*

$$0 \longrightarrow H_H^j(Z_p, Z_{p-1}) \longrightarrow H_H^j(Z_p, Z_{m_0-1}) \longrightarrow H_H^j(Z_{p-1}, Z_{m_0-1}) \longrightarrow 0. \quad (7.16)$$

Restriction induces an isomorphism

$$H_H^j(\Lambda, Z_{m-1}) \longrightarrow H_H^j(Z_m, Z_{m-1}) \quad \text{if } j < a_{m+1}. \quad (7.17)$$

Proof. First use $H = R$, inducting on p . By (7.12), the earlier interval module is generated over the ambient classes w, e by chosen extensions of its bottoms x_j , $m_0 \leq j < p$. The connecting homomorphism sends such a bottom into the new crossing in degree $a_j + 1 < a_p$, where the crossing is zero. It is linear over w, e , hence vanishes on the whole earlier module. This proves (7.16) for R , and also justifies the inductive generation assertion.

For G , Lemma 7.4 gives the same induction with the bottom generators and, at even stages j , the chosen top generators of degree $a_j + n$. Bottoms again have zero connecting image. For a top, the only potentially relevant degree is $a_j + n + 1$. If $d > 2$, then $n + 1 = d + 2 < 2d$, so this degree is less than a_p . If $d = 2$, equality can occur only when $j = p - 1$ is even. In this case p is odd, and the target is its one-dimensional bottom G -crossing. Restriction to the corresponding reflection bottom is injective: both bottom Thom classes restrict to the nonzero bottom class. The reflection connecting map has already been proved zero. Naturality therefore kills the G -connecting map in this last case as well. This proves (7.16) in every dimension $n \geq 3$.

We pass to the global pair degree by degree. Every singular chain in the Borel construction is finite and has bounded round energy, since energy descends to a continuous invariant function. Thus

$$H_j^H(\Lambda, Z_{m-1}) = \varinjlim_{p \geq m} H_j^H(Z_p, Z_{m-1}).$$

Universal coefficients over the field $\mathbb{F}_2$ identify its dual with the inverse limit of the degree- j cohomology groups. The local crossing groups vanish below a_p , which tends to infinity. Sequence (7.16) consequently stabilizes in every degree. For $j < a_{m+1}$, no subsequent crossing contributes, proving (7.17). All limits in this argument are taken in a fixed degree; the total cohomology module is the graded direct sum. $\square$

Since Z_0 retracts equivariantly to M , the triple exact sequence identifies $H_H^*(Z_p, Z_0)$ with $H_H^*(Z_p, M)$, and likewise $H_H^*(\Lambda, Z_0)$ with $H_H^*(\Lambda, M)$. Thus the first interval filtration computes precisely the global groups relative to constants used below.

Choose the relative extension

$$\tilde{\zeta}_m \in H_G^{a_m}(\Lambda, Z_{m-1})$$

of the bottom Thom class. Map it to $H_G^{a_m}(\Lambda, M)$, and call the resulting class ζ_m . Let $X_m = \text{res}_R \zeta_m$. For even m , choose y_m as in Lemma 7.4. Since

$$n = d + 1 < 2d, \tag{7.18}$$

it has a unique relative extension in degree $a_m + n$. Call its image in $H_G^*(\Lambda, M)$ Y_m . Restriction and cup by e commute with the relative extensions; both have local value $e x_m$. Uniqueness in (7.17) therefore gives

$$\text{res}_R Y_m = e X_m \quad (m \text{ even}). \tag{7.19}$$

Similarly, $w^n \text{res}_R \tilde{\zeta}_m$ has local value zero by (7.11). Its offset is $n < 2d$, so uniqueness gives

$$w^n X_m = 0. \tag{7.20}$$

The products $w^k X_m, e w^k X_m$, $0 \leq k \leq d$, now form a basis of B_∞ . To see independence, restrict a finite relation to the lowest round stage occurring in it: later classes vanish there, while the remaining leading terms form the local basis (7.12). Repeat with successive stages. For spanning, in each fixed degree the interval short exact sequences give precisely the sum of the local dimensions, and only finitely many stages contribute. Equation (7.20) and multiplication by e show that w^n annihilates the entire relative module.

7.6 Odd-stage torsion

The remaining full-group relation follows from one integration calculation. This will also keep the dimension-three case within the strict extension range.

Lemma 7.6 (Torsion of odd bottoms). *For every odd m and every $n \geq 3$,*

$$U^n \zeta_m = 0.$$

When $n = 3$, the supported relative lift also satisfies

$$w^3 \tilde{\zeta}_m = 0, \quad U(w^2 + U) \tilde{\zeta}_m = 0.$$

Proof. The recurrence (7.14) implies

$$A_i^2 + A_{i-1} A_{i+1} = U^i \quad (i \geq 1). \quad (7.21)$$

Indeed, substitution of the recurrence makes the left side for $i \geq 2$ equal to U times the left side for $i - 1$; the initial value is $w^2 + A_2 = U$.

At an odd crossing, (7.15) gives

$$U A_d x_m = (A_{d+2} + w A_{d+1}) x_m = 0.$$

The class $A_d x_m$ is nonzero: on restriction to R , it becomes $w^d x_m$, which is nonzero in (7.12). The relevant part of the standard-plane Gysin sequence is

$$H_R^{a_m+d+1}(Z_m, Z_{m-1}) \xrightarrow{\int} H_G^{a_m+d}(Z_m, Z_{m-1}) \xrightarrow{U} H_G^{a_m+d+2}(Z_m, Z_{m-1}).$$

Its source is one-dimensional, spanned by $e x_m$. Since its image contains the nonzero class $A_d x_m$, exactness over $\mathbb{F}_2$ forces

$$\int e x_m = A_d x_m. \quad (7.22)$$

Apply the same integration map to $e \operatorname{res}_R \tilde{\zeta}_m$ in the pair (Λ, Z_{m-1}) . Naturality and (7.22) identify its restriction to the crossing with $A_d x_m$. Both sides have offset $d < 2d$, so (7.17) gives

$$\int (e \operatorname{res}_R \tilde{\zeta}_m) = A_d \tilde{\zeta}_m.$$

The next map in Gysin is multiplication by U , hence

$$U A_d \tilde{\zeta}_m = 0. \quad (7.23)$$

Also $A_{d+1} \tilde{\zeta}_m = 0$, because its local value is zero and $d + 1 < 2d$. Multiplying (7.21) at $i = d$ by $U \tilde{\zeta}_m$ therefore gives

$$U^{d+1} \tilde{\zeta}_m = U A_d^2 \tilde{\zeta}_m + U A_{d-1} A_{d+1} \tilde{\zeta}_m = 0.$$

Its image in $H_G^*(\Lambda, M)$ is $U^n \zeta_m = 0$.

For $n = 3$, one has $d = 2$, $A_2 = w^2 + U$, and $A_3 = w^3$. The two intermediate global relations are exactly

$$w^3 \tilde{\zeta}_m = 0, \quad U(w^2 + U) \tilde{\zeta}_m = 0.$$

Their proof used uniqueness only at offsets two and three, both strictly less than the next-bottom gap four. It did not extend an offset-four relation across that gap. $\square$

The basis construction, (7.19), and Lemma 7.6 complete the proof of Theorem 7.2. For later use, the degree intervals for the bottom family are $[a_m, a_m + d]$, and those for the top family are $[a_m + n, a_m + n + d]$. Intervals within either family are disjoint, since their starting degrees differ by $2d$; a bottom and a top can have the same degree. Thus restrictions of tops will be counted modulo all restricted bottoms. The next sections compare those global restrictions with the extra classes present at a finite cut.

8 A finite-cut defect bound

Throughout this section we assume that there are finitely many prime geodesic images. Our objective is to compare a sublevel with its half-translation fixed locus while retaining the information lost by localization. The comparison has two degrees: the total equivariant degree upstairs and the degree of a root loop downstairs. An exact-sequence calculation will bound their discrepancy, summed over all degree cutoffs, by the number of critical events.

Write

$$B_a^l = H_R^l(X_a, M), \quad B_\infty^l = H_R^l(\Lambda, M).$$

At a regular cut these reflection groups have finite total dimension: Theorem 3.7 identifies each reflection crossing with a finite sum of ordinary one-orientation crossing groups, and there are only finitely many crossings below a finite cut. All sublevels in the statements are open. At regular endpoints, the deformations of Lemmas 3.2 and 3.3 identify them, naturally under inclusion and fixed restriction, with closed regular polygon models. We use those compact models in the localization argument.

Lemma 8.1 (Finite-pair localization). *Let $2a > 0$ be regular. Then a is regular, and restriction to the half-translation fixed locus induces an isomorphism*

$$H_K^*(X_{2a}, M)[v^{-1}] \longrightarrow H_K^*(X_{2a}^\tau, M)[v^{-1}].$$

It is an actual isomorphism in every sufficiently high degree, with the threshold allowed to depend on a . The same localization assertion holds for crossing pairs and respects their individual prime labels.

For an ambient crossing at the m -th iterate of a prime, its actual K -group has uniformly bounded dimension in every degree, is zero below $I_m - A$, and maps isomorphically to its stabilized fixed target above $I_m + A$, after increasing the uniform constant A . For odd m that target is zero; for even m its root labels satisfy $|l - I_{m/2}| \leq A$.

Proof. A critical root doubles to a critical loop, so regularity of $2a$ implies regularity of a . Doubling identifies X_a with X_{2a}^τ as an R -space. Since τ acts trivially there, a product universal model and the field Künneth formula give

$$H_K^J(X_{2a}^\tau, M) = \bigoplus_{l \leq J} v^{J-l} B_a^l. \quad (8.1)$$

The same identification holds for fixed crossing pairs.

Choose a common even polygon model through the cut, and double its compact regular sublevel along an invariant collar. Denote the resulting closed smooth K -manifold by $\mathcal{D}$. Inclusion of the original sublevel and the fold map make its restriction arrow to the fixed locus a retract of the corresponding arrow for $\mathcal{D}$. The fixed submanifold $\mathcal{D}^\tau$ has an invariant open tubular neighborhood O_0 which retracts equivariantly onto it. The compact complement of O_0 is covered by finitely many invariant open sets $O_1, \dots, O_s$, each admitting an equivariant map to K/H_i with $\tau \notin H_i$. Indeed, use a small stabilizer-invariant neighborhood disjoint from its translates by other cosets, and then saturate it.

The possible groups H_i are 1 , $R = \langle r \rangle$, and $\langle \tau r \rangle$. On each, $U = v(v + w)$ restricts to zero. Thus U lifts to $H_K^2(\mathcal{D}, O_i)$. Cup these lifts with a class relative to O_0 . The relative cup product for the open cover has target relative to $\bigcup_{i=0}^s O_i = \mathcal{D}$, so U^s annihilates $H_K^*(\mathcal{D}, O_0)$. The exact sequence of this pair proves localization at U . Passing to the retract proves it on the original sublevel. The exact sequences for a pair of regular sublevels, and for the pair relative to M , give the corresponding assertions for pairs.

The class w is zero on each ambient local block, because that block is induced from the $\langle \tau \rangle$ -pair of one orientation. It is therefore nilpotent on each finite-cut relative group

by induction over the crossing exact sequences. The same argument applies to the fixed reflection groups and to crossing pairs. On these modules localization at U is equivalent to localization at v : after inverting v , the factor $1 + w/v$ in $U = v^2(1 + w/v)$ is invertible; after inverting U , the element $(v + w)/U$ is an inverse for v .

Apply the sign-line Gysin sequence of Proposition 7.1 to one orientation of a local block. The total ordinary rank bound and index support in Theorem 3.7 imply the claimed per-degree bound, vanishing below the index window, and that multiplication by v is an isomorphism above it. The fixed local group has the product description in (8.1). Its root support is within a fixed distance of $I_{m/2}$, and

$$|I_m - 2I_{m/2}| \leq n.$$

Consequently both source and target have already stabilized above $I_m + A$, with a uniform enlarged A . The localized restriction isomorphism is then an actual isomorphism there: clear a power of v and invert the intervening, already invertible multiplication maps. For an odd iterate the fixed block is empty, so its stabilized source is zero. Labelwise naturality follows from the common full/fixed lens in Theorem 3.7.

Finally, the sign-line Gysin sequence from K to R and the finite total dimension of B_{2a} make multiplication by v an isomorphism on the ambient relative group in all sufficiently high degrees. The fixed group has the same property by (8.1). The same clearing argument upgrades localization to actual restriction in high degrees at this cut. $\square$

For a paired regular cut $(2a, a)$, put

$$\mathcal{V}_a = \bigoplus_l B_a^l, \quad A_J = H_K^J(X_{2a}, M).$$

Dropping the powers v^{J-l} in (8.1) defines the normalized actual restriction $\epsilon_J : A_J \rightarrow \mathcal{V}_a$. Set

$$E_J = \text{im } \epsilon_J, \quad t_J = \dim \ker \epsilon_J.$$

The image E_J lies in labels at most J ; it need not be graded by root label, and it need not equal the stabilized target. In these normalized coordinates, multiplication by v leaves a vector unchanged, whereas multiplication by w applies the root w to each component and raises its label by one.

Let P_j denote projection to root labels at most j . We record three defects: the dimension missing from the projected low root degrees, the dimension of actual-image vectors supported entirely above the cutoff, and the dimension of actual classes invisible on the fixed locus. They are, respectively, $D_{j,q}$, $H_{j,J}$, and t_J . The potential also includes the preceding-degree deficit $D_{j,2j-1}$; this compensates for the degree shift in the connecting homomorphism. Define

$$\begin{aligned} D_{j,q} &= \dim P_j \mathcal{V}_a - \dim P_j E_q, \\ H_{j,J} &= \dim(E_J \cap \ker P_j), \quad J = 2j, \\ \Phi_j(a) &= D_{j,2j} + D_{j,2j-1} + H_{j,2j} + t_{2j}, \quad j \geq 1. \end{aligned} \tag{8.2}$$

All four terms are nonnegative. Lemma 8.1 shows that they vanish for sufficiently large j at any fixed cut.

Theorem 8.2 (Finite-cut defect bound). *There is a constant C such that, whenever $2a > 0$ is regular,*

$$\sum_{j \geq 1} \Phi_j(a) \leq C(1 + a). \tag{8.3}$$

Proof. Grow the ambient cut through its critical events, using primes for the groups just after an event. Let C^q be its actual local K -group and $\mathcal{W}$ its total fixed root reflection group. The comparison maps fit into the following diagram, whose rows are exact:

$$\begin{array}{ccccccccc}
A_{q-1} & \longrightarrow & C^q & \longrightarrow & A'_q & \longrightarrow & A_q & \longrightarrow & C^{q+1} \\
\epsilon_{q-1} \downarrow & & \lambda_q \downarrow & & \epsilon'_q \downarrow & & \epsilon_q \downarrow & & \lambda_{q+1} \downarrow \\
\mathcal{V} & \xrightarrow{\delta} & \mathcal{W} & \xrightarrow{f} & \mathcal{V}' & \xrightarrow{i} & \mathcal{V} & \xrightarrow{\delta} & \mathcal{W}.
\end{array} \tag{8.4}$$

The bottom row is the fixed exact sequence after inverting v and normalizing powers consistently in neighboring degrees. Its maps f, i preserve root labels, and δ raises a root label by one. In particular, the repeated terms denote the total graded groups in an exact sequence, not an assertion of ungraded periodicity.

Fix $j \geq 1$ and put $J = 2j$. Call this cutoff *far* from the event if every participating iterate satisfies $|I_m - 2j| > C_0$, for a sufficiently large fixed C_0 . Lemma 8.1 and the index comparison then give

$$\mathcal{W}^j = \mathcal{W}^{j+1} = 0, \quad \lambda_{J-1}, \lambda_J, \lambda_{J+1} \text{ injective with image } P_j \mathcal{W}. \tag{8.5}$$

Indeed, the blocks below the window are already in their actual stable range and have all root labels below j ; those above it have zero actual source in these three degrees and no low root labels.

Write $P = P_j$, $L = P\mathcal{V}$, $L' = P\mathcal{V}'$, and

$$A_0 = f(P\mathcal{W}), \quad r = \text{rank}(\delta|_L), \quad r_q = \text{rank}(\delta|_{PE_q}) \quad (q = J-1, J).$$

There is no connecting term across the projection edge, so low-label exactness gives

$$\dim P\mathcal{W} = \dim A_0 + r, \quad \dim L' - \dim L = \dim A_0 - r. \tag{8.6}$$

We next justify the projected-image calculation, including mixed low and high components. For $z \in E_q$, commutativity and (8.5) place δz in $P\mathcal{W}$. Also $\delta(Pz)$ is low, since $\mathcal{W}^{j+1} = 0$, while $\delta((1-P)z)$ has labels greater than j . These disjoint label ranges imply

$$\delta((1-P)z) = 0, \quad \delta z = \delta(Pz).$$

Given $x \in PE_q \cap \ker \delta$, choose $z \in E_q$ projecting to x and an actual representative in A_q . Its actual connecting class restricts to zero. The target λ_{q+1} is injective, so actual exactness lifts the representative to A'_q . Projecting its fixed restriction proves

$$i(PE'_q) = PE_q \cap \ker \delta.$$

The kernel of this map is exactly A_0 : stable exactness gives the inclusion in A_0 , and every element of $P\mathcal{W}$ has an actual local lift. Therefore

$$D'_{j,q} - D_{j,q} = -(r - r_q), \quad q = J-1, J. \tag{8.7}$$

The actual connecting ranks are precisely r_q , again by target injectivity. Hence

$$\dim A'_J - \dim A_J = \dim P\mathcal{W} - r_{J-1} - r_J.$$

Using $H_{j,J} + t_J = \dim A_J - \dim PE_J$ and (8.6) gives

$$(H_{j,J} + t_J)' - (H_{j,J} + t_J) = r - r_{J-1}.$$

Together with (8.7), this proves the exact change

$$\Phi'_j - \Phi_j = -(r - r_J) \leq 0. \quad (8.8)$$

The preceding-degree deficit cancels the possible increase in the high-image and kernel terms. This is why all four terms were retained.

For a cutoff which is not far, local restriction need not be injective. Nevertheless the change is uniformly bounded. Choose M_0 bounding $\dim \mathcal{W}$ and all the relevant actual local dimensions. Since i preserves labels, $i(PE'_q) \subset PE_q$. Its kernel has dimension at most M_0 by stable exactness. Its cokernel is bounded using the surjections

$$A_q / \text{im } A'_q \longrightarrow E_q / iE'_q \longrightarrow PE_q / i(PE'_q).$$

The first source has dimension at most $\dim C^{q+1} \leq M_0$. Thus $|\dim PE'_q - \dim PE_q| \leq M_0$. The same kernel and cokernel argument bounds $|\dim L' - \dim L|$ by M_0 , and actual exactness bounds $|\dim A'_J - \dim A_J|$ by M_0 . Since

$$\Phi_j = 2 \dim L - 2 \dim PE_J - \dim PE_{J-1} + \dim A_J,$$

we obtain $|\Phi'_j - \Phi_j| \leq 6M_0$.

Each prime contributes at most one iterate to a given positive event. The finite prime list therefore bounds the number of participating blocks, and only a bounded number of integers j lie in their fixed index windows. Thus every event increases the sum of the potentials by at most a fixed constant. The number of events below $2a$ is at most $\sum_i \lfloor 2a/\ell_i \rfloor$. The potentials start at zero below the first positive event and are unchanged on no-critical strips. Their finite support at each cut permits summation of (8.8) and the exceptional bounds, proving (8.3). $\square$

The estimate controls all degree cutoffs at once. In the next section, a single persistent extra reflection class will produce quadratically many units of this linear budget. Degree zero, which is not included in (8.2), will be bounded separately by ordinary death.

9 A uniform bound for transient reflection classes

We now bound the total dimension of reflection-equivariant classes at a finite cut modulo those coming from the full loop space. Ordinary death forces such a class, away from finitely many index bands, to propagate through a long string of powers of w . The finite-cut budget rules out that string. Bounds in the remaining degrees then give a bound on the total dimension.

Throughout this section $n \geq 3$, $d = n - 1$, and we assume that there are only finitely many prime closed-geodesic images. Let $\ell_i > 0$ and $\Delta_i \geq 0$ be their prime lengths and mean indices, as in Theorem 3.7; thus

$$|I_{i,m} - m\Delta_i| \leq n, \quad \beta_i = \Delta_i/\ell_i. \quad (9.1)$$

For every positive regular cut a , retain

$$\begin{aligned} B_a^j &= H_R^j(X_a, M), & B_\infty^j &= H_R^j(\Lambda, M), \\ Q_a^j &= \text{coker}(B_\infty^j \longrightarrow B_a^j), & T(a) &= \sum_{j \geq 0} \dim Q_a^j. \end{aligned}$$

All coefficients are $\mathbb{F}_2$. The local crossing bounds imply that $\bigoplus_j B_a^j$ is finite-dimensional at each such cut.

Theorem 9.1 (Bounded total transients). *Under the finite-prime-image assumption, there is a constant C , independent of the positive regular cut a , such that*

$$T(a) \leq C. \quad (9.2)$$

We use four previously established inputs. Theorem 6.1 provides a single filling increment D in ordinary homology relative to M , independent of both degree and cut. Theorem 3.7 bounds local ranks and their index supports. Theorem 7.2 gives $w^n B_\infty = 0$. Finally, Theorem 8.2 controls the discrepancy between actual ambient degree and root degree. The ordinary rank and resonance statements used below are those of Proposition 5.1; write $\alpha > 0$ for its resonance slope. Constants in this section may depend on the metric, n , and the finite list of prime images, but never on a or j .

9.1 Degree gaps and multiplication by w

Let

$$\mathcal{S} = \{0, \alpha, \beta_i : \text{prime labels } i\} = \{s_0 < s_1 < \cdots < s_N\}$$

be the list of distinct slopes. Thus $s_0 = 0$ and $N \geq 1$. For a fixed margin $c > 0$ and consecutive slopes $s < s'$, define the integer degree gap

$$\mathcal{I}_{s,s'}(a) = \{j \in \mathbb{Z}_{\geq 0} : sa + c < j < s'a - c\}.$$

Since α is among the slopes, each gap lies entirely below or entirely above αa . We call these gaps present and absent, respectively.

Lemma 9.2 (Comparison on degree gaps). *There is a fixed margin c such that, for every positive regular a and every consecutive pair $s < s'$ in $\mathcal{S}$, the following statements hold.*

1. *In a present gap, $B_\infty^j \rightarrow B_a^j$ is injective for every $j \in \mathcal{I}_{s,s'}(a)$.*
2. *In an absent gap, $B_\infty^j \rightarrow B_a^j$ is zero for every $j \in \mathcal{I}_{s,s'}(a)$.*
3. *Whenever $j, j+1 \in \mathcal{I}_{s,s'}(a)$, multiplication by w induces an isomorphism $Q_a^j \rightarrow Q_a^{j+1}$.*

Moreover $B_a^j = 0$ for $j > s_N a + c$.

Proof. We first prove the ordinary comparison away from fixed-width bands. Choose a regular endpoint $a_+ \in (a + D, a + D + 1)$. A local group at an event $m\ell_i \in [a, a_+]$ can contribute in degree j or its neighboring degrees only if

$$|j - I_{i,m}| \leq C_{\text{loc}}.$$

By (9.1), this implies

$$|j - \beta_i a| \leq C_{\text{loc}} + n + \beta_i(D + 1). \quad (9.3)$$

The right-hand side is bounded uniformly over the finite prime list. Thus, outside fixed-width bands about the $\beta_i a$, the ordinary map

$$H_j(X_a, M) \longrightarrow H_j(X_{a_+}, M)$$

is injective, by the crossing exact sequences. A class in its source that is zero globally is already zero in X_{a+D} by Theorem 6.1. Injectivity therefore makes the map $H_j(X_a, M) \rightarrow H_j(\Lambda, M)$ injective. Field duality gives surjectivity of ordinary cohomology restriction.

In degrees above a fixed low threshold, the ordinary groups relative to M identify with the absolute groups, by the evaluation retraction and the cohomology of the constant sphere. In these degrees Proposition 5.1 gives global rank at most one. Its resonance estimate implies

that every nonzero global class restricts nontrivially to X_a when $j < \alpha a - c_0$, and restricts to zero when $j > \alpha a + c_0$, for a fixed c_0 . Together with the preceding surjectivity, this proves

$$\begin{cases} H^j(\Lambda, M) \longrightarrow H^j(X_a, M) \text{ is an isomorphism,} & j < \alpha a - c_0, \\ H^j(X_a, M) = 0, & j > \alpha a + c_0, \end{cases} \quad (9.4)$$

provided j also lies outside the fixed low-degree interval and the bands in (9.3). When the global group is zero, surjectivity gives the first assertion as an isomorphism between zero groups. Increase c_0 to absorb all these restrictions.

Consider a present gap and put $P_a^j = H_R^j(\Lambda, X_a)$ for this proof. Ordinary triple exactness and (9.4) give $H^j(\Lambda, X_a) = 0$ whenever the needed two neighboring ordinary restrictions are isomorphisms. The sign-line Gysin sequence of Proposition 7.1 consequently makes $w : P_a^j \rightarrow P_a^{j+1}$ an isomorphism in the corresponding interior. After allowing n further degrees below, every element of P_a^j is divisible by w^n there. Its image in B_∞^j is zero, since $w^n B_\infty = 0$. Applying this at j and $j+1$ to the triple (Λ, X_a, M) gives a natural short exact sequence

$$0 \longrightarrow B_\infty^j \longrightarrow B_a^j \longrightarrow P_a^{j+1} \longrightarrow 0. \quad (9.5)$$

The quotient identification commutes with w . It gives both the claimed injection and the isomorphisms on Q_a in a present gap.

In an absent gap, (9.4) and sign Gysin make $w : B_a^j \rightarrow B_a^{j+1}$ an isomorphism in the interior. Allowing n further degrees above makes w^n injective there. The image of B_∞^j is killed by that power, so it is zero. Hence $Q_a^j = B_a^j$ and the claimed isomorphisms also hold in an absent gap.

Choose the margin c larger than $c_0 + 2n + 6$ and the fixed low-degree threshold. Then all neighboring degrees and all powers just used lie in the required untrimmed ordinary ranges whenever the statements of the lemma apply. This is one fixed choice for all cuts and gaps. Finally, any reflection local contribution below cut a has

$$j \leq I_{i,m} + C_{\text{loc}} \leq \beta_i a + n + C_{\text{loc}} \quad (m\ell_i < a).$$

The relative groups start at zero below the first event. The local crossing sequences therefore give $B_a^j = 0$ above $s_N a + c$, after one final increase of c . $\square$

9.2 The finite-cut budget excludes long strings

Lemma 9.3 (No transients inside a long degree gap). *For all sufficiently large regular cuts a with $2a$ regular, $Q_a^j = 0$ on every gap $\mathcal{I}_{s,s'}(a)$ of Lemma 9.2.*

Proof. Fix consecutive slopes $s < s'$ and write

$$A = sa, \quad r = (s' - s)a.$$

There is a fixed $\delta > 0$ such that $r \geq \delta a$ for every such gap, since the list of distinct slopes is finite. Suppose that some Q_a^j in this gap is nonzero. The isomorphisms in Lemma 9.2 make it nonzero throughout the integer gap. For large a , choose

$$l_0 = \lceil A + 0.10r \rceil$$

and a class $y_{l_0} \in B_a^{l_0}$ with nonzero image in $Q_a^{l_0}$. Set $y_l = w^{l-l_0} y_{l_0}$ through $l \leq \lfloor A + 0.90r \rfloor$. Every resulting quotient class is nonzero. In particular, with

$$\mathcal{L} = \mathbb{Z} \cap [A + 0.15r, A + 0.20r], \quad q = \lfloor 0.50r \rfloor,$$

the classes y_l , $l \in \mathcal{L}$, are independent in $\bigoplus_j B_a^j$, as are their powers $w^q y_l$. Indeed they occupy distinct root degrees, and their quotient classes remain nonzero between degrees $A + 0.65r - O(1)$ and $A + 0.70r$. Figure 2 displays the intervals used below.

At the ambient cut $2a$, let E_J and P_k be the actual normalized restriction image and root-degree projection from Theorem 8.2. Recall its nonnegative terms

$$D_{k,J} = \dim P_k \left(\bigoplus_l B_a^l \right) - \dim P_k E_J, \quad H_{k,J} = \dim(E_J \cap \ker P_k).$$

In particular that theorem gives

$$\sum_{k \geq 1} (D_{k,2k} + H_{k,2k}) \leq C(1 + a). \quad (9.6)$$

Average $D_{k,2k}$ over the integers $k \in [A + 0.30r, A + 0.35r]$. There are a fixed positive multiple of r such integers for large r , so some k satisfies $D_{k,2k} \leq C_1$, independently of a , because $r \geq \delta a$. The space

$$W_0 = \text{span}\{y_l : l \in \mathcal{L}\}$$

lies in the root degrees at most k . Thus

$$\dim(W_0 \cap P_k E_{2k}) \geq \dim W_0 - D_{k,2k} \geq c_1 r$$

for a fixed $c_1 > 0$ and sufficiently large r .

Choose a basis $z_1, \dots, z_\nu$ of this intersection and actual ambient lifts

$$u_\mu \in H_K^{2k}(X_{2a}, M).$$

The normalized fixed restriction of u_μ has the form $z_\mu + c_\mu$, where every root label in c_μ is greater than k . Multiply the actual classes by w^q . The selected terms $w^q z_\mu$ remain independent, since w^q is injective on W_0 . Their labels are at most $A + 0.70r$, whereas the other terms have labels greater than

$$k + q \geq A + 0.80r - 1$$

or vanish. The latter terms cannot cancel any of the selected ones. Moreover all labels of these normalized images exceed $A + 0.64r$ for sufficiently large r .

For any integer $k' \in [A + 0.61r, A + 0.64r]$, there is enough actual degree to multiply further by a nonnegative power of v , since

$$2k + q \leq 2A + 1.20r < 2A + 1.22r \leq 2k'. \quad (9.7)$$

The classes

$$v^{2k' - (2k + q)} w^q u_\mu \in H_K^{2k'}(X_{2a}, M)$$

have unchanged normalized fixed images: multiplication by v changes the actual degree and leaves the root coefficients unchanged. Multiplication by w raises each root label by one, as used above. Thus the images are independent and lie entirely above root cutoff k' , proving

$$H_{k',2k'} \geq c_1 r \quad \text{for every integer } k' \in [A + 0.61r, A + 0.64r].$$

There are at least $c_2 r$ such cutoffs for large r . The left side of (9.6) is therefore at least $c_1 c_2 r^2 \geq c_1 c_2 \delta^2 a^2$, contradicting its linear upper bound. This proves the assertion for the chosen gap. The finite number of gaps allows one common lower threshold for a . $\square$

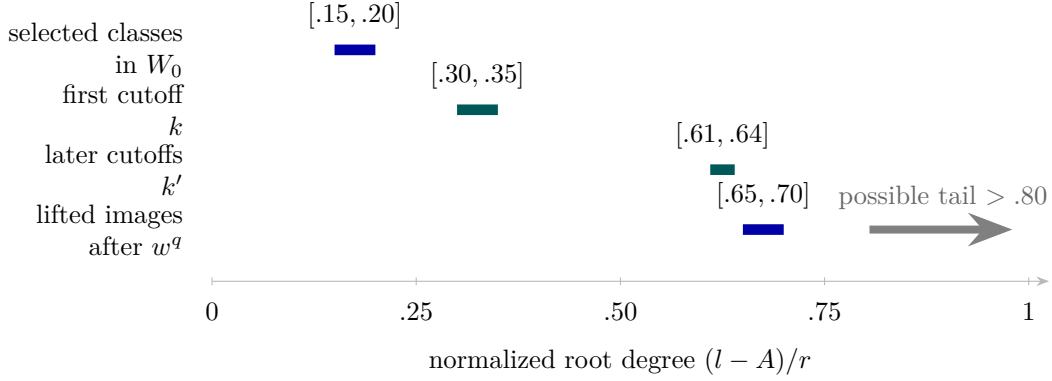


Figure 2: The degree intervals in Lemma 9.3, normalized by the gap width r ; the drawing suppresses rounding errors of order r^{-1} . Multiplication by w^q , with $q/r \approx .50$, moves the selected root degrees above every later cutoff k' while keeping them separate from the other terms. Multiplication by v then raises actual degree without changing root labels; the inequality $2k + q < 2k'$ provides the required degree room.

9.3 Bounding the remaining degrees

Proof of Theorem 9.1. First let a be large with $2a$ regular. By Lemma 9.3, transient classes can occur only within distance c of one of the finitely many degrees $s_i a$. Lemma 9.2 excludes degrees above the last such band. Thus only a bounded number of integer degrees can contribute. We now bound the dimension in each of them uniformly.

For high degrees, the reflection local ranks of Theorem 3.7 give

$$\dim B_a^j \leq C \sum_i \#\{m \geq 1 : m\ell_i < a, |j - m\Delta_i| \leq C_{\text{loc}} + n\}. \quad (9.8)$$

This follows by summing the crossing exact-sequence bounds, starting with the zero relative group below the first event. If $\Delta_i > 0$, the number in braces is bounded by $1 + 2(C_{\text{loc}} + n)/\Delta_i$, up to an integer rounding constant. If $\Delta_i = 0$, it is zero once $j > C_{\text{loc}} + n$. The finite prime list therefore gives a uniform bound on $\dim B_a^j$ in every sufficiently high degree.

The fixed low-degree interval needs a different argument, which also accounts for all zero-mean-index iterates. Choose a regular $a_+ \in (a + D, a + D + 1)$. By ordinary death,

$$\ker(H_j(X_a, M) \rightarrow H_j(\Lambda, M)) = \ker(H_j(X_a, M) \rightarrow H_j(X_{a_+}, M)). \quad (9.9)$$

The kernel on the right is bounded in dimension by the sum of local ordinary ranks in this interval of cut values of width at most $D + 1$. The number of its events is at most

$$\sum_i \left(2 + \frac{D + 1}{\ell_i}\right),$$

independently of a . Uniform local ranks bound the kernel, and the global-image rank is bounded by the fixed global degree rank. Consequently $\dim H^j(X_a, M)$ is bounded uniformly in a for each fixed low degree j .

The sign-line Gysin sequence gives

$$\dim B_a^j \leq \dim B_a^{j-1} + \dim H^j(X_a, M), \quad B_a^{-1} = 0.$$

Induction, including degree zero, bounds B_a^j uniformly throughout any fixed low-degree interval. Combining this with (9.8), every degree in the finitely many remaining bands has

uniformly bounded dimension. Since the number of such degrees is also bounded, their sum bounds $T(a)$ as asserted.

Cuts a in a bounded interval have only finitely many crossing stages, so they satisfy a common bound as well. Finally, if a is regular but $2a$ is not, choose a nearby a' in the same no-critical interval such that $2a'$ is regular. The critical values and their halves are locally finite. The no-critical restriction isomorphism between the root sublevels is natural with respect to global restriction, and hence preserves both B_a and its global image. In particular $T(a) = T(a')$. This proves (9.2) for every positive regular cut. $\square$

We have bounded the total excess dimension at each regular cut without assuming monotonicity as the cut varies. The next section proves monotonicity under doubling. Together, these statements will force an interval of maximal transient dimension on which the activation of a new global class is incompatible with the local evaluation-cup relation.

10 Doubling and the first activation

Theorem 9.1 bounds the total number of reflection classes which do not come from the whole loop space. We now compare that number at a cut and its double. Equality in this comparison will require each visible odd-stage bottom class to have a visible evaluation-cup partner. At the first appearance of such a bottom, local evaluation vanishing prevents its partner from appearing. This gives the final contradiction.

Retain the finite-prime assumption and write

$$V(a) = \sum_j \text{rank}(B_\infty^j \longrightarrow B_a^j), \quad T(a) = \sum_j \dim B_a^j - V(a).$$

Both sums are finite at a regular cut. We first record the full-group rank calculation that makes the doubling comparison exact.

Lemma 10.1 (Free rank and torsion at a finite cut). *Suppose that $2L > 0$ is regular, and put $H = \bigoplus_j H_G^j(X_{2L}, M)$. This is a finitely generated graded $\mathbb{F}_2[U]$ -module. Its torsion submodule $\mathcal{T}$ is finite-dimensional and annihilated by a power of U . If*

$$t = \dim(\mathcal{T}/U\mathcal{T}) = \dim \ker(U|_H), \quad \rho = \text{rank}_{\mathbb{F}_2[U]}(H/\mathcal{T}),$$

then

$$\sum_j \dim B_{2L}^j = \rho + 2t, \quad \rho = \sum_j \dim B_L^j. \quad (10.1)$$

Proof. The standard-plane Gysin sequence in Proposition 7.1 embeds H/UH into $\bigoplus_j B_{2L}^j$, which has finite dimension. Lift a finite homogeneous basis of H/UH . These lifts generate H : subtract their contributions modulo U , divide the remainder by U , and induct down the nonnegative degree. This is an induction decreasing degree by two at each step.

If a polynomial $p(U)$ has nonzero constant term, it acts injectively on H . Indeed, the lowest nonzero homogeneous component of a nonzero element survives in its product with $p(U)$. Factoring an annihilating polynomial as $U^s p(U)$ therefore shows that every torsion element is killed by a power of U . The ring $\mathbb{F}_2[U]$ is Noetherian, so $\mathcal{T}$ has finitely many generators and is killed by one common power of U . It is consequently finite-dimensional. Rank-nullity on $\mathcal{T}$ gives

$$\dim \ker(U|_H) = \dim \ker(U|_{\mathcal{T}}) = \dim(\mathcal{T}/U\mathcal{T}).$$

Moreover $UH \cap \mathcal{T} = U\mathcal{T}$, since Uh torsion implies that h is torsion.

Set $\mathcal{F} = H/\mathcal{T}$. Multiplication by U is injective on $\mathcal{F}$. Lift a homogeneous basis of $\mathcal{F}/U\mathcal{F}$. Degree induction again proves generation. These lifts are free: in a homogeneous relation cancel the least power of U and reduce modulo U . Independence of the selected basis excludes the terms with that least power, a contradiction. Thus $\mathcal{F}$ has a homogeneous free basis of size ρ , and

$$\dim(H/UH) = \rho + t.$$

Summing the standard-plane Gysin sequence over degrees gives the first identity in (10.1), since its two contributions are H/UH and a shifted copy of $\ker U$.

For sufficiently large D , the torsion no longer contributes in degrees $D, D-1$. The dimensions of those two free degrees add to ρ , one contribution from each free generator according to parity. The weight-two-plane Gysin sequence gives

$$\rho = \dim H_G^D(X_{2L}, M) + \dim H_G^{D-1}(X_{2L}, M) = \dim H_K^D(X_{2L}, M).$$

By Lemma 8.1 and (8.1), the last dimension equals $\sum_j \dim B_L^j$ in high degree. This proves the second identity without requiring surjectivity of full-group restriction to the fixed locus. $\square$

Use the global basis from Theorem 7.2:

$$\xi_{m,k} = w^k X_m, \quad \eta_{m,k} = e\xi_{m,k}, \quad m \geq 1, \quad 0 \leq k \leq d.$$

At a regular cut a , let $b_{m,k}(a)$ be one if $\xi_{m,k}$ restricts nontrivially and zero otherwise. Let $h_{m,k}(a)$ be one if the restriction of $\eta_{m,k}$ is independent modulo all bottom restrictions in its degree, and zero otherwise. Degrees within the bottom family are distinct, as are degrees within the top family: their consecutive starting degrees differ by $2d$, while each family at a stage spans an interval of width d . Therefore

$$V(a) = \sum_{m,k} (b_{m,k}(a) + h_{m,k}(a)), \quad h_{m,k}(a) \leq b_{m,k}(a). \quad (10.2)$$

The inequality follows from $\eta_{m,k} = e\xi_{m,k}$. Write $V_o(a)$ and $V_e(a)$ for the contributions of odd and even stages m to this sum. Stage parity here refers to the round-filtration label m .

Theorem 10.2 (Doubling and its equality condition). *For every regular $2L > 0$,*

$$T(2L) \geq T(L). \quad (10.3)$$

If equality holds, then

$$h_{m,k}(2L) = b_{m,k}(2L) \quad (m \text{ odd}, \quad 0 \leq k \leq d). \quad (10.4)$$

Proof. Use the notation of Lemma 10.1, and put

$$N = \sum_{m \text{ odd}, k} b_{m,k}(2L).$$

The selected odd bottoms have global G -equivariant U -torsion lifts by Theorem 7.2. Their restrictions to H/UH are independent, since that quotient injects into reflection cohomology and their bottom restrictions are independent. They thus occupy an N -dimensional subspace of $\mathcal{T}/U\mathcal{T}$. Write

$$t = N + J_0, \quad J_0 \geq 0. \quad (10.5)$$

The selected even bottoms and tops also have global full-group lifts. Their classes modulo U are jointly independent with the selected odd bottoms: this follows from their reflection

restrictions and the rule that a top is counted only modulo all bottoms in its degree. In the projection

$$H/UH \longrightarrow \mathcal{F}/U\mathcal{F},$$

the kernel is $\mathcal{T}/U\mathcal{T}$, of dimension $N + J_0$. The odd span already occupies N of these dimensions. Hence the image of the even span has dimension at least $V_e(2L) - J_0$.

Choose a maximal homogeneous independent family among these even projections; its size is at least $V_e(2L) - J_0$. Multiply the corresponding global lifts by powers of U to put them in two common large degrees $D, D - 1$, according to parity. Independence persists in $\mathcal{F}$: from a relation in one of the common degrees, cancel the least power of U and reduce modulo U , then repeat if necessary. Thus the sum of the ambient full-group global-image ranks in these two degrees is at least $V_e(2L) - J_0$.

We compare these images with the fixed locus, including the image-level Gysin maps. The subspace X_{2L}^τ is G -invariant. Under its identification with X_L , rotations act with doubled parameter and reflection acts as before. In sufficiently high degree the map

$$H_G^q(X_{2L}, M) \longrightarrow H_G^q(X_{2L}^\tau, M)$$

is injective. Indeed, compose it with the injective fixed G -to- K restriction. The same composite factors through the ambient G -to- K injection and the high actual K restriction isomorphism of Lemma 8.1.

For $H' = G, K$, set

$$I_{H'}^q = \text{im}[H_{H'}^q(\Lambda^\tau, M) \longrightarrow H_{H'}^q(X_{2L}^\tau, M)].$$

Naturality places the ambient global images inside these fixed global images. Apply the weight-two-plane Gysin sequence to the fixed global and fixed finite pairs. Its global push is surjective, so on images the finite push gives a surjection

$$\pi : I_K^D \longrightarrow I_G^{D-1}.$$

The finite G -to- K injection identifies I_G^D with a subspace of $\ker \pi$. Consequently

$$\dim I_G^D + \dim I_G^{D-1} \leq \dim I_K^D.$$

On the fixed K -spaces the global restriction is the coefficientwise map $B_\infty[v] \rightarrow B_L[v]$. Since B_L has finite total dimension, $\dim I_K^D = V(L)$ for sufficiently large D . Choose D large enough for all the preceding comparisons. We have proved

$$V(L) \geq V_e(2L) - J_0. \tag{10.6}$$

Subtract global-image dimensions from (10.1) and use (10.5)–(10.6):

$$\begin{aligned} T(2L) - T(L) &= 2(N + J_0) - V_o(2L) - V_e(2L) + V(L) \\ &\geq J_0 + 2N - V_o(2L) \\ &= J_0 + \sum_{m \text{ odd}, k} (b_{m,k}(2L) - h_{m,k}(2L)) \geq 0. \end{aligned}$$

All summands in the final line are nonnegative. Equality therefore forces (10.4) for each individual odd bottom. $\square$

Completion of the proof of Theorem 1.1 for $n \geq 3$. Suppose that there are only finitely many prime geodesic images. By Theorem 9.1, T is a bounded nonnegative integer on regular

positive cuts. It is constant on every no-critical interval, so its maximum is attained on a nonempty bounded open regular interval $I \subset (0, \infty)$.

Every regular point $x \in 2^k I$ has regular successive halves: a critical half would double to a critical point. Iterating (10.3) therefore gives $T(x) \geq T(x/2^k)$, which is already the global maximum. Thus every regular point in every expanded interval attains that maximum. For $k \geq 1$, equality holds between x and $x/2$, so (10.4) holds at every such x . This uses monotonicity under doubling only; no monotonicity between arbitrary cuts is asserted.

Take k large and choose regular $x_1 < x_2$ in the lower and upper quarters of $2^k I$. The interval of degrees strictly between αx_1 and αx_2 , after removing the fixed endpoint margins of Lemma 9.2, has length tending to infinity. Remove also the fixed-width bands about every $\beta_i x_1$ and $\beta_i x_2$. They remove a bounded total number of possible degrees. The odd bottom degrees

$$a_m = (2m - 1)d, \quad m \text{ odd},$$

form an arithmetic progression of spacing $4d$. For sufficiently large k , one such degree $j = a_m$ remains, beyond the fixed low-degree range. Lemma 9.2 makes $\xi_{m,0}$ invisible at x_1 and visible at x_2 . At x_1 the degree lies in an absent gap, where the global image is zero, or above the last band, where the sublevel group is zero. At x_2 it lies in a present gap where global reflection restriction is injective.

A global class nonzero at a smaller sublevel cannot become zero at a larger one, since restriction to the smaller sublevel factors through the larger. There are only finitely many events between x_1 and x_2 . Choose the first event at which $\xi_{m,0}$ becomes nonzero, and tight regular cuts $x_- < x_+$ around it, still inside $2^k I$. The exact sequence of the crossing puts its post-event restriction in the image of

$$H_R^j(X_{x_+}, X_{x_-}) \longrightarrow B_{x_+}^j,$$

because its pre-event restriction is zero. By Theorem 3.7, cup with e vanishes on this whole reflection crossing, including simultaneous prime labels. Naturality of the cup action therefore gives

$$\eta_{m,0}|_{X_{x_+}} = e \xi_{m,0}|_{X_{x_+}} = 0.$$

Hence $b_{m,0}(x_+) = 1$ while $h_{m,0}(x_+) = 0$. Since m is odd, this contradicts (10.4) at the regular point x_+ of the expanded interval. The finite-prime-image assumption is impossible. $\square$

11 Finite spherical covers

Proof of Corollary 1.2. Let $\pi : \widetilde{M} \rightarrow M$ be a finite smooth covering of degree r , where $\widetilde{M}$ is diffeomorphic to S^n . Equip $\widetilde{M}$ with the pullback of the given metric. The map π is then a local isometry, so it sends every nonconstant closed geodesic upstairs to a nonconstant closed geodesic downstairs. Taking the least positive period of the latter does not change its image.

We show that at most r upstairs prime images project to any one downstairs prime image. Choose an oriented unit-speed representative $\sigma : \mathbb{R} \rightarrow M$ of that image, of least period $L > 0$, and put $p = \sigma(0)$. For each $q \in \pi^{-1}(p)$ there is a unique lift of σ starting at q , with initial velocity $(d\pi_q)^{-1}\dot{\sigma}(0)$. Lifting one period permutes the finite fiber $\pi^{-1}(p)$; lifting the reversed path gives the inverse permutation. After $m \leq r$ periods a lift therefore returns to its initial point. Its velocity also returns, because the downstairs velocity returns and $d\pi_q$ is invertible. The lift is a closed geodesic. If m is the first return of that fiber point, its least period is mL : any period upstairs projects to a period of σ and hence is an integer multiple of L .

By Lemma 3.4, unit-speed geodesics with the same image agree after a parameter shift and possibly reversal. Consequently any upstairs prime geodesic whose projected image is the chosen one can be parametrized so that its projection is exactly σ . It is then one of the lifts

starting at the r points above p . This proves the finite-fiber bound. The argument permits self-intersections and does not require the covering to be regular; phase or reversal can only identify some of these finitely many images.

Theorem 1.1 gives infinitely many upstairs prime images for the pullback metric. Since every downstairs prime image has at most r preimages in this set, there are infinitely many downstairs prime images as well. $\square$

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